

ASVH-METH-001

Engineering Methodology for the NHS Ambient Scribing Runtime Verification Harness

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- ASVH-AUTH-001
- ASVH-SPEC-001
- ASVH-TRACE-001
- ASVH-001 (Reference Implementation Report)

Executive Summary

The adoption of Artificial Intelligence (AI) within healthcare is accelerating, with AI-enabled ambient scribing systems offering significant opportunities to reduce clinician administrative burden while improving the quality and consistency of clinical documentation. NHS England has recognised both the potential and the risks associated with these technologies through the publication of guidance defining the governance, clinical safety and organisational responsibilities required for their safe deployment across health and care settings.

Although this guidance establishes what organisations should achieve, it is intentionally written for human interpretation rather than deterministic software implementation. As a result, a significant engineering challenge remains: how can narrative governance guidance be transformed into precise, traceable and implementation-independent engineering artefacts capable of supporting deterministic runtime verification?

The **Ambient Scribing Verification Harness (ASVH) Engineering Methodology** addresses this challenge by defining a structured engineering process that systematically transforms published NHS England guidance into runtime-verifiable requirements suitable for implementation within an independent Runtime Authority. Rather than creating new governance requirements or redefining existing policy, the methodology preserves the intent of the published guidance while expressing that intent through a disciplined sequence of engineering activities.

The methodology introduces a repeatable engineering lifecycle that begins with the analysis and decomposition of NHS guidance into discrete operational statements. These statements are classified according to their engineering significance and progressively transformed into operational requirements, runtime-verifiable conditions, Runtime Context definitions and deterministic Runtime Rules. Throughout this process, complete traceability is maintained between the original guidance and every derived engineering artefact, ensuring that implementation remains transparent, auditable and reproducible.

The methodology is implementation independent. It does not prescribe a particular ambient scribing supplier, Electronic Patient Record (EPR), technology platform or deployment architecture. Instead, it defines the engineering discipline required to derive deterministic runtime behaviour while preserving clear separation between governance, requirements engineering, runtime specification, software architecture and implementation.

This publication defines:

- the engineering principles governing runtime requirements derivation;
- the methodology for analysing and decomposing NHS England guidance;
- the classification of operational statements into engineering artefacts;
- the derivation of runtime-verifiable requirements;
- the transformation of requirements into Runtime Context and Runtime Rules;
- the maintenance of end-to-end engineering traceability; and
- the engineering outputs required to support deterministic runtime verification.

The methodology forms the foundation of the wider ASVH engineering framework. Its outputs are realised through the companion publications **ASVH-TRACE-001** (Guidance Traceability Matrix), **ASVH-SPEC-001** (Runtime Authority Specification), **ASVH-ARCH-001** (Runtime Architecture) and **ASVH-001** (Reference Implementation). Together, these publications establish a complete engineering pathway from published NHS guidance through to deterministic runtime evaluation and evidential Authority Receipt generation.

The ASVH Engineering Methodology does **not** define clinical policy, governance decisions, software architecture or implementation technologies. Those responsibilities remain within the relevant companion publications and existing NHS governance frameworks. Instead, this document defines the engineering discipline required to transform narrative operational guidance into implementation-independent, deterministic runtime verification.

The principal contribution of this publication is the introduction of a structured and repeatable engineering methodology that bridges the gap between policy and execution. By combining guidance decomposition, runtime requirements engineering, deterministic rule derivation and complete engineering traceability, the methodology provides a robust foundation upon which independent Runtime Authority implementations can be consistently designed, implemented and verified.

Engineering Methodology at a Glance

Engineering Challenge

NHS England guidance for AI-enabled ambient scribing establishes the governance, clinical safety and operational expectations for the deployment of ambient scribing technologies within healthcare organisations. However, these publications are written for human interpretation and organisational implementation rather than deterministic software execution.

For Runtime Authority to evaluate execution admissibility consistently, narrative guidance must first be transformed into precise engineering artefacts that can be traced, implemented and verified. This requires a disciplined engineering methodology that preserves the intent of the original guidance while expressing it in a form suitable for deterministic runtime evaluation.

The ASVH Engineering Methodology defines that transformation process.

Engineering Contribution

The methodology introduces a structured engineering lifecycle that progressively converts published NHS guidance into implementation-independent runtime verification.

Rather than creating new governance requirements or modifying existing NHS policy, the methodology preserves the intent of the published guidance while transforming it through a series of documented engineering activities. Each stage produces clearly defined outputs that become the inputs to the next stage, maintaining complete traceability throughout the engineering lifecycle.

The result is a repeatable engineering process capable of producing deterministic Runtime Contexts, Runtime Rules and Runtime Authority behaviour that remain directly traceable to their originating guidance.

Engineering Lifecycle

Engineering Stage	Primary Output
NHS England Guidance Analysis	Operational Statements
Guidance Classification	Engineering Categories
Operational Requirement Engineering	Operational Requirements
Runtime Requirement Engineering	Runtime Requirements
Runtime Context Definition	Runtime Context Model
Runtime Rule Derivation	Deterministic Runtime Rules
Traceability Engineering	Guidance-to-Implementation Traceability
Runtime Architecture	Implementation-Independent Reference Architecture
Reference Implementation	Executable Runtime Authority
Runtime Verification	Authority Receipts and Engineering Evidence

Core Engineering Principles

The methodology is founded upon six engineering principles:

- **Preservation of Guidance Intent** – Engineering artefacts accurately reflect the intent of the originating NHS guidance.
 - **Deterministic Engineering** – Derived runtime behaviour produces repeatable and reproducible outcomes.
 - **Complete Traceability** – Every engineering artefact can be traced back to its originating guidance statement.
 - **Implementation Independence** – The methodology remains independent of vendors, software platforms and deployment technologies.
 - **Separation of Engineering Responsibilities** – Guidance analysis, requirements engineering, runtime specification, architecture and implementation remain distinct engineering activities.
 - **Evidential Accountability** – Every engineering decision is supported by documented provenance and traceable engineering outputs.
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Position within the ASVH Engineering Framework

This publication provides the engineering methodology underpinning the entire ASVH engineering framework.

Publication	Primary Responsibility
ASVH-METH-001	Runtime Requirements Engineering Methodology
ASVH-TRACE-001	Guidance-to-Implementation Traceability
ASVH-SPEC-001	Runtime Authority Behaviour Specification
ASVH-ARCH-001	Implementation-Independent Runtime Architecture
ASVH-001	Reference Implementation and Verification

Together, these publications define a complete engineering pathway from published NHS guidance through to deterministic runtime verification and evidential Authority Receipt generation.

Scope of this Publication

This publication defines the engineering methodology used to derive implementation-independent Runtime Authority behaviour from published NHS England guidance.

It defines:

- guidance analysis and decomposition;
- operational and runtime requirements engineering;
- runtime context derivation;
- deterministic runtime rule derivation;
- engineering traceability; and
- methodology outputs supporting the wider ASVH engineering framework.

It does **not** define software architecture, runtime behaviour, implementation technologies or healthcare governance policy. These responsibilities are addressed by the companion ASVH publications.

Chapter 1 – Introduction

1.1 Purpose

The adoption of AI-enabled ambient scribing technologies is changing the way clinical documentation is created across healthcare. By capturing clinical conversations and generating draft documentation, these systems have the potential to reduce administrative burden while supporting more efficient clinical workflows.

Recognising both the opportunities and the associated risks, NHS England has published guidance defining the governance, clinical safety and operational expectations for the deployment of ambient scribing systems within health and care organisations. These publications establish the conditions under which AI-assisted documentation may be introduced into clinical practice while preserving patient safety, clinical accountability and organisational governance.

Although this guidance provides comprehensive operational direction, it is intentionally expressed as narrative policy rather than deterministic engineering artefacts. Consequently, it does not define how individual guidance statements should be transformed into runtime-verifiable requirements suitable for implementation within independent Runtime Authority systems.

This document addresses that engineering challenge by defining a structured methodology for systematically transforming published NHS England guidance into implementation-independent engineering artefacts capable of supporting deterministic runtime verification.

Rather than introducing new governance requirements, the methodology provides a disciplined engineering process that preserves the intent of published guidance while enabling consistent implementation across different technology platforms and healthcare environments.

1.2 Objectives

The objectives of the ASVH Engineering Methodology are to:

- establish a repeatable engineering process for analysing NHS England guidance;
- define a systematic approach to deriving operational and runtime requirements;
- provide a structured methodology for constructing deterministic Runtime Contexts and Runtime Rules;
- maintain complete engineering traceability from published guidance through to implementation;
- preserve implementation independence throughout the engineering lifecycle; and

- support the development of verifiable Runtime Authority implementations capable of deterministic execution-time evaluation.

Collectively, these objectives provide the engineering discipline required to transform narrative operational guidance into deterministic runtime verification while maintaining transparency, reproducibility and evidential accountability.

1.3 Intended Audience

This publication is intended for:

- enterprise architects responsible for healthcare system design;
- software engineers implementing Runtime Authority capabilities;
- clinical safety officers and digital transformation teams;
- AI-enabled ambient scribing suppliers;
- NHS organisations evaluating runtime verification approaches;
- systems engineers responsible for requirements engineering and implementation traceability.

The methodology assumes familiarity with systems engineering concepts but does not require prior knowledge of the ASVH engineering framework.

1.4 Relationship to Companion Publications

The methodology presented within this document forms the engineering foundation of the wider ASVH publication suite.

Its outputs are progressively realised by the accompanying publications:

- **ASVH-TRACE-001**, which maintains complete traceability between NHS guidance and derived engineering artefacts.
- **ASVH-SPEC-001**, which defines the deterministic Runtime Rules and Runtime Authority behaviour derived from the methodology.
- **ASVH-ARCH-001**, which defines the implementation-independent Runtime Architecture supporting those engineering artefacts.
- **ASVH-001**, which demonstrates the methodology through a working reference implementation and verification evidence.

Together these publications provide a complete engineering pathway from published NHS guidance through to deterministic runtime verification and evidential Authority Receipt generation.

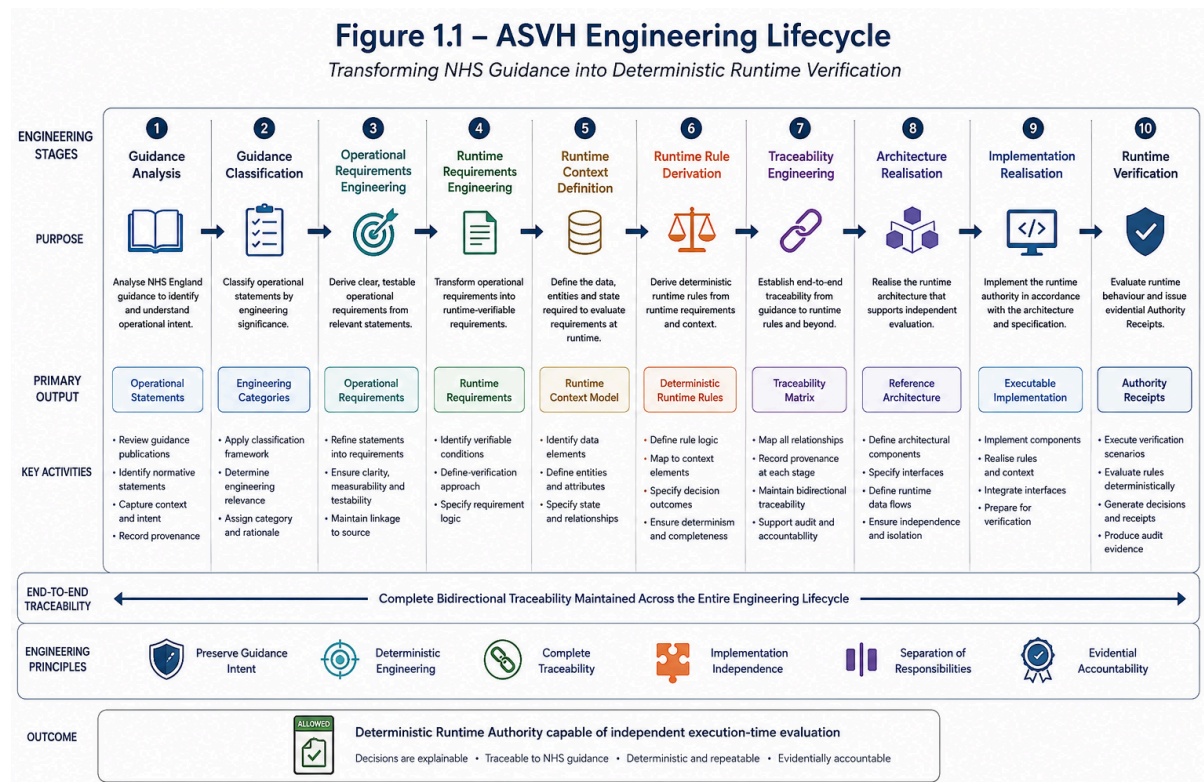


Figure 1.1 – ASVH Engineering Lifecycle

Figure 1.1 illustrates the complete engineering lifecycle defined by the ASVH Engineering Methodology for transforming published NHS England guidance into deterministic Runtime Authority behaviour. The figure demonstrates that the methodology is not a software development process, but a structured systems engineering approach that progressively converts narrative operational guidance into implementation-independent engineering artefacts suitable for deterministic runtime verification.

The lifecycle begins with the analysis of NHS England guidance, where operational intent is identified and individual guidance statements are classified according to their engineering significance. These classified statements are then transformed into measurable operational requirements before being further refined into runtime-verifiable requirements that define the information required immediately before execution.

From these runtime requirements, a formal Runtime Context model is constructed, providing the structured operational data required for deterministic evaluation. This Runtime Context forms the basis for deriving deterministic Runtime Rules that express each runtime condition in a precise, repeatable and implementation-independent manner. Throughout this process, complete engineering traceability is maintained to preserve the relationship between the originating NHS guidance and every derived engineering artefact.

The resulting Runtime Rules are subsequently realised through the companion ASVH publications, including the Runtime Architecture, Runtime Authority Specification and Reference Implementation, culminating in deterministic runtime verification and the generation of immutable Authority Receipts as engineering evidence.

The figure also highlights the engineering principles that govern every stage of the lifecycle, including preservation of guidance intent, deterministic engineering, complete traceability, implementation independence, separation of engineering responsibilities and evidential accountability. These principles ensure that identical source guidance, interpreted through the same engineering methodology, produces consistent and reproducible engineering outputs irrespective of implementation technology or deployment environment.

Figure 1.1 establishes the conceptual foundation for the remainder of this publication. Subsequent chapters examine each stage of the engineering lifecycle in detail, describing the processes, engineering artefacts and traceability mechanisms required to transform narrative healthcare guidance into deterministic runtime verification capable of supporting independent Runtime Authority implementations.

The remainder of this publication examines each stage of the engineering lifecycle shown in Figure 1.1, describing how operational guidance is systematically transformed into deterministic runtime verification while preserving implementation independence and complete engineering traceability.

Chapter Summary

This chapter has introduced the purpose, objectives and intended audience of the ASVH Engineering Methodology. It has established the engineering problem addressed by the methodology and positioned the publication within the broader ASVH engineering framework.

The following chapter defines the scope of the methodology and the engineering boundaries within which the subsequent requirements engineering process is performed.

Chapter 2 – Scope

2.1 Scope

The ASVH Engineering Methodology defines a structured systems engineering process for transforming published NHS England guidance into implementation-independent engineering artefacts capable of supporting deterministic Runtime Authority implementations.

The methodology establishes the engineering activities required to analyse narrative operational guidance, derive operational and runtime requirements, construct Runtime Context definitions, develop deterministic Runtime Rules and maintain complete engineering traceability throughout the engineering lifecycle.

The methodology is intended to provide a repeatable engineering process rather than a software implementation methodology. It defines the engineering outputs required to support Runtime Authority implementations while remaining independent of programming language, software architecture, deployment model or technology platform.

Accordingly, this publication describes **how engineering artefacts are derived**, rather than **how Runtime Authority software is implemented**.

2.2 Engineering Scope

The methodology encompasses the following engineering activities:

- analysis of NHS England guidance relevant to AI-enabled ambient scribing;
- identification and classification of operational statements;
- derivation of operational requirements;
- engineering of runtime-verifiable requirements;
- definition of Runtime Context elements;
- derivation of deterministic Runtime Rules;
- maintenance of end-to-end engineering traceability;
- production of engineering artefacts supporting architecture, specification and implementation.

Each activity produces documented outputs that provide the inputs to subsequent stages of the engineering lifecycle.

2.3 Out of Scope

The methodology does not define:

- NHS governance policy or organisational operating procedures;
- clinical decision-making or clinical safety assessment;
- implementation-specific Runtime Rules;
- software architecture or deployment topology;
- programming languages, frameworks or technology platforms;
- supplier-specific ambient scribing implementations;
- Electronic Patient Record design or integration;
- operational management of healthcare systems.

These responsibilities are addressed by existing NHS governance frameworks or by the companion ASVH publications.

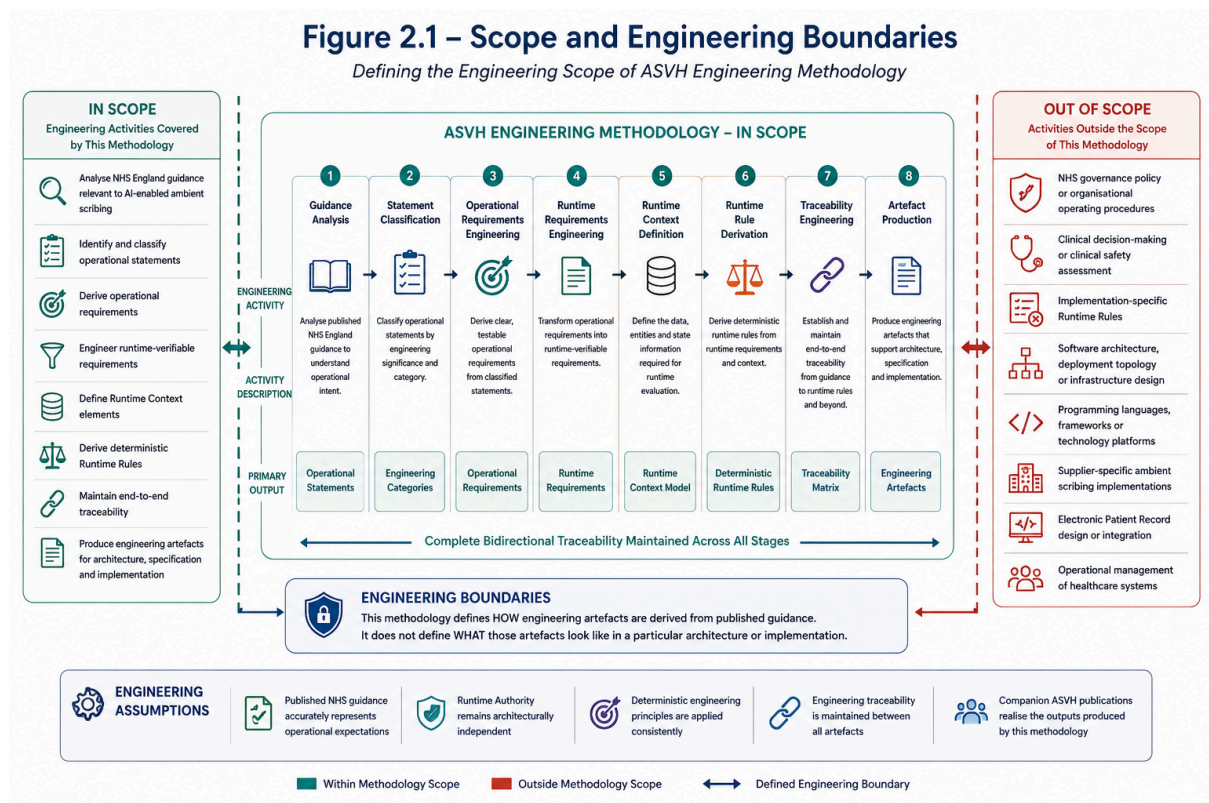


Figure 2.1 – Scope and Engineering Boundaries

Description

Figure 2.1 illustrates the engineering scope of the ASVH Engineering Methodology and defines the architectural boundaries within which the methodology operates. The figure distinguishes the engineering activities performed by the methodology from the wider governance, architectural and implementation responsibilities that remain outside its scope.

Within the defined engineering boundary, the methodology transforms published NHS England guidance through a structured sequence of engineering activities including guidance analysis, operational statement classification, operational and runtime requirements engineering, Runtime Context definition, Runtime Rule derivation, engineering traceability and engineering artefact production. Each activity produces documented outputs that become the inputs to the subsequent stage, maintaining complete bidirectional traceability throughout the engineering lifecycle.

The figure also identifies the activities that are intentionally excluded from the methodology. Governance policy, clinical decision-making, software architecture, implementation technologies, supplier-specific integrations and operational healthcare management remain separate engineering or organisational responsibilities addressed by existing NHS frameworks and the companion ASVH publications. By explicitly defining these exclusions, the methodology preserves clear separation between engineering process and implementation.

The engineering boundary shown within the figure emphasises that this publication defines **how engineering artefacts are systematically derived from published guidance**, rather than prescribing how those artefacts are realised within a particular Runtime Authority implementation. This distinction preserves implementation independence while allowing the methodology to support multiple architectural approaches and technology platforms.

Supporting the methodology are a set of engineering assumptions that underpin every stage of the lifecycle. These include preservation of the intent of published NHS guidance, architectural independence of Runtime Authority, consistent application of deterministic engineering principles, complete engineering traceability and the progressive realisation of engineering outputs by the companion ASVH publications.

Figure 2.1 establishes the engineering boundaries that govern the remainder of this publication. Subsequent chapters examine each activity contained within the methodology while maintaining the separation of responsibilities illustrated by the defined engineering scope. This disciplined separation ensures that the ASVH Engineering Methodology remains a repeatable, implementation-independent systems engineering process capable of transforming narrative operational guidance into deterministic runtime verification artefacts.

2.4 Engineering Assumptions

The methodology assumes that:

- published NHS guidance accurately represents the intended operational expectations;
- Runtime Authority remains architecturally independent of governance and implementation;
- deterministic engineering principles are applied consistently throughout the lifecycle;

- engineering traceability is maintained between all derived artefacts;
- companion publications realise the outputs produced by this methodology.

These assumptions establish the engineering boundaries within which the methodology operates.

2.5 Relationship to Companion Publications

The engineering methodology defines the process by which engineering artefacts are produced.

Those artefacts are subsequently realised by the companion publications:

Publication	Primary Responsibility
ASVH-TRACE-001	Maintains traceability between guidance and engineering artefacts.
ASVH-SPEC-001	Defines Runtime Rules and Runtime Authority behaviour.
ASVH-ARCH-001	Defines the implementation-independent Runtime Architecture.
ASVH-001	Demonstrates the methodology through a working reference implementation.

Together these publications provide a complete engineering framework supporting deterministic runtime verification for AI-enabled ambient scribing.

Chapter Summary

This chapter has defined the scope and engineering boundaries of the ASVH Engineering Methodology. It has distinguished the engineering activities addressed by this publication from the architectural, specification and implementation responsibilities defined elsewhere within the ASVH engineering framework.

The following chapter introduces the engineering lifecycle that underpins the methodology and describes the sequential activities through which published NHS England guidance is transformed into deterministic runtime verification.

3. Engineering Lifecycle

3.1 Overview

The ASVH Engineering Methodology follows a structured lifecycle that transforms published operational guidance into an executable reference implementation capable of deterministic runtime verification.

Each phase produces a discrete engineering artefact that contributes to the next stage of development. This progression ensures that every runtime behaviour demonstrated by the reference implementation can be traced to documented engineering decisions derived from the original NHS England guidance.

The methodology deliberately separates analysis, design, implementation and verification activities. Each phase has a defined purpose, documented outputs and explicit relationships to subsequent artefacts.

The resulting engineering chain provides transparency, repeatability and explainability throughout the development process.

3.2 Engineering Lifecycle

The engineering lifecycle introduced in Chapter 1 is expanded in this chapter through a detailed description of each engineering activity and the relationships between the engineering artefacts produced at every stage. Figure 3.1 illustrates the complete engineering lifecycle that governs the transformation of published NHS England guidance into deterministic runtime verification.

Figure 3.1 – ASVH Engineering Lifecycle



Figure 3.1 – ASVH Engineering Lifecycle

The lifecycle illustrates that the executable reference implementation is not the starting point of the engineering process. Rather, it is the final expression of a sequence of documented engineering activities beginning with published NHS England guidance.

3.3 Lifecycle Stages

Stage 1 — Source Guidance

The methodology begins with published NHS England guidance relating to the safe use of AI-enabled ambient scribing products.

At this stage, no engineering decisions are made.

The guidance is treated as the authoritative source material from which operational expectations will subsequently be derived.

The methodology deliberately avoids extending, modifying or reinterpreting the published guidance beyond the engineering transformations described within this document.

Stage 2 — Engineering Analysis

The guidance is examined to identify statements that describe expected operational behaviour immediately preceding clinical consequence formation.

Engineering analysis distinguishes between:

- descriptive background information;
- organisational governance responsibilities;
- operational expectations;
- runtime-relevant conditions.

Only runtime-relevant operational expectations progress to the next stage of the methodology.

Stage 3 — Requirement Extraction

Operational expectations identified during engineering analysis are expressed as structured engineering requirements.

Each extracted requirement captures a single observable runtime characteristic that can subsequently be evaluated by deterministic logic.

Requirement extraction intentionally avoids implementation decisions.

At this stage, the methodology defines **what must be verified**, not **how verification will occur**.

Stage 4 — Runtime Context Definition

Each extracted requirement is mapped to one or more Runtime Context elements.

Runtime Context defines the operational information that must be available immediately before execution can be evaluated.

Examples include:

- consultation state;
- documentation review status;
- execution destination;
- operational environment;
- intended workflow.

Runtime Context therefore establishes the observable state required for deterministic runtime verification.

Stage 5 — Runtime Rule Derivation

Runtime Rules are derived directly from documented Runtime Context.

Each rule evaluates a single deterministic execution condition.

The methodology intentionally favours simple, independently explainable rules over complex composite logic.

This approach improves traceability, simplifies verification and enables individual rule outcomes to be reconstructed within the Authority Receipt.

Stage 6 — Traceability Construction

Traceability links every Runtime Rule back through the engineering process to its originating operational requirement and ultimately to the published NHS England guidance.

This traceability is preserved throughout the publication series and reference implementation.

The objective is to ensure that every observable runtime behaviour can be explained through documented engineering evidence rather than implementation-specific assumptions.

Stage 7 — Reference Architecture

Once Runtime Rules have been defined, the methodology identifies the architectural location at which deterministic runtime verification should occur.

For ASVH, this resulted in the identification of the **Execution Bind Point** as the architectural boundary immediately preceding consequence formation.

The Reference Architecture documents the responsibilities of each architectural component while preserving separation between clinical review, runtime verification and execution.

Stage 8 — Verification Scenario Development

Verification scenarios are designed before implementation.

Each scenario represents a controlled operational condition used to demonstrate deterministic runtime behaviour.

The scenario set includes:

- expected operational behaviour;
- incomplete review states;
- invalid execution contexts;
- operational failures;
- boundary conditions.

The objective is to demonstrate predictable runtime outcomes across representative execution states.

Stage 9 — Reference Implementation

Only after completion of the preceding engineering activities is the executable reference implementation developed.

The implementation is intentionally presented as **one implementation of the methodology**, rather than the methodology itself.

Its purpose is to execute the documented Runtime Rules, evaluate Runtime Context, produce deterministic decisions and generate Authority Receipts in accordance with the preceding engineering artefacts.

Stage 10 — Runtime Evaluation and Evidence

During execution, the Runtime Authority evaluates the Runtime Context against the derived Runtime Rules.

The resulting decision, supporting evidence and execution metadata are recorded within an Authority Receipt.

The Authority Receipt represents the final engineering artefact produced by the methodology and provides a structured record of the runtime verification process.

3.4 Engineering Characteristics

The lifecycle has been designed to exhibit five fundamental engineering characteristics.

Characteristic	Description
Traceable	Every implementation element can be traced to published guidance.
Deterministic	Identical Runtime Context produces identical outcomes.
Explainable	Runtime decisions can be reconstructed from documented engineering artefacts.
Implementation Independent	The methodology is independent of any specific technology stack or vendor product.
Repeatable	The engineering process can be applied consistently to future operational guidance.

These characteristics collectively ensure that the methodology supports transparent engineering, facilitates independent review and enables consistent reproduction of the reference implementation.

4. Source Analysis and Guidance Decomposition

4.1 Introduction

The ASVH engineering methodology begins with the systematic analysis of published NHS England guidance relating to the safe deployment and operation of AI-enabled ambient scribing products.

The objective of this stage is not to reinterpret guidance or create new operational requirements. Rather, it is to identify those elements of the published guidance that describe conditions relevant to runtime execution and to transform them into engineering artefacts capable of deterministic evaluation.

This distinction is fundamental to the methodology. Guidance remains the authoritative source of operational intent, while engineering provides the structured process through which that intent is represented as executable runtime behaviour.

4.2 Engineering Objectives

The source analysis phase was designed to achieve five objectives.

1. Preserve the original intent of the published guidance.
2. Identify operational expectations that influence execution at runtime.
3. Distinguish runtime verification requirements from broader governance responsibilities.
4. Maintain complete traceability between source guidance and engineering artefacts.
5. Produce engineering outputs suitable for deterministic evaluation.

These objectives ensure that the engineering process remains transparent, repeatable and independently reviewable.

4.3 Guidance Analysis Process

The methodology applies a structured decomposition process to each relevant statement within the source guidance.

Rather than attempting to engineer directly from narrative text, the guidance is progressively transformed through a series of increasingly precise engineering representations.

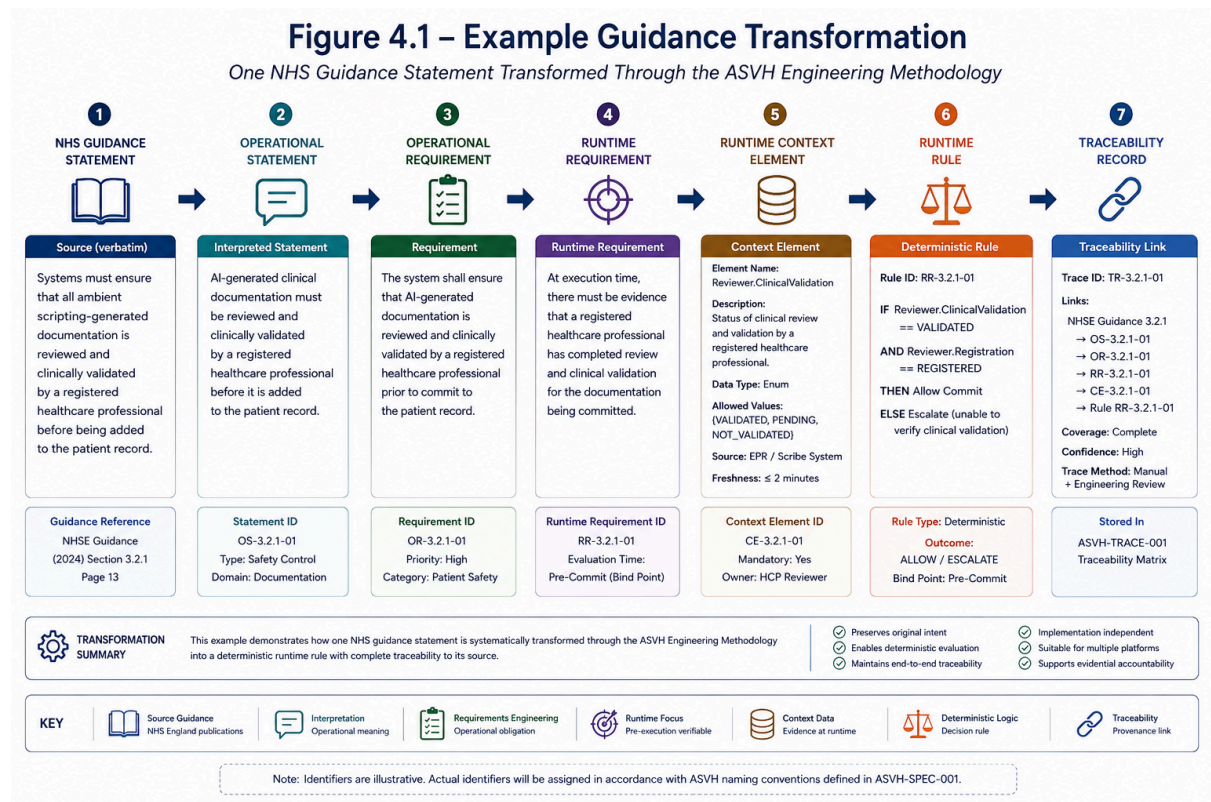


Figure 4.1 – Example Guidance Transformation

Figure 4.1 demonstrates how a single statement from published NHS England guidance is systematically transformed into deterministic engineering artefacts through the ASVH Engineering Methodology. Unlike the engineering lifecycle presented in Chapter 3, which describes the overall methodology, this figure follows one representative guidance statement through every stage of engineering transformation.

The process begins with an NHS England guidance statement expressed in narrative form. This statement is first interpreted as an operational statement, preserving the original intent while expressing the expected operational behaviour in a form suitable for engineering analysis. From this interpretation, a formal operational requirement is derived that defines a measurable obligation capable of subsequent verification.

The operational requirement is then refined into a runtime requirement describing the conditions that must be satisfied immediately before institutional consequence formation. These runtime requirements are used to define the corresponding Runtime Context elements that capture the operational information required during execution-time evaluation.

Using the Runtime Context as its foundation, a deterministic Runtime Rule is derived that expresses the runtime requirement in an implementation-independent form suitable for evaluation by an independent Runtime Authority. The resulting rule defines the conditions under which execution may be permitted, escalated or refused while preserving deterministic behaviour.

Throughout the transformation process, every engineering artefact is linked through an explicit traceability record. This record preserves the relationship between the originating NHS guidance statement, the operational interpretation, derived requirements, Runtime Context elements and Runtime Rule, providing complete engineering provenance from policy through to deterministic runtime evaluation.

Figure 4.1 illustrates the principal objective of the ASVH Engineering Methodology: preserving the intent of published guidance while progressively expressing that intent through increasingly precise engineering artefacts without introducing implementation-specific assumptions. The resulting Runtime Rule remains directly traceable to its originating guidance statement, supporting transparency, reproducibility, deterministic evaluation and evidential accountability.

This transformation process provides the engineering foundation for the companion ASVH publications, where the derived Runtime Rules, Runtime Context definitions and traceability relationships are formalised within the Runtime Authority Specification, Runtime Architecture and Reference Implementation. By demonstrating the complete engineering pathway for a single guidance statement, the figure establishes a repeatable model that can be applied consistently across the wider body of NHS England ambient scribing guidance.

4.4 Operational Statement Identification

The first engineering activity is the identification of operational statements contained within the published guidance.

Operational statements describe behaviours, expectations or conditions that influence the safe use of ambient scribing systems.

During analysis, guidance is examined to distinguish operational statements from supporting explanatory material.

Examples of supporting material include:

- background information;
- implementation advice;
- contextual discussion;
- policy rationale.

While valuable for understanding the broader context, these elements are not directly transformed into runtime verification artefacts.

Instead, the methodology focuses on statements that describe observable operational behaviour immediately preceding consequence formation.

4.5 Operational Requirement Definition

Each operational statement is translated into a structured engineering requirement.

The purpose of this activity is to remove narrative ambiguity while preserving the intent of the original guidance.

An operational requirement describes **what condition must exist** before execution may proceed.

At this stage no implementation assumptions are introduced.

The methodology intentionally avoids prescribing software architecture, implementation technology or evaluation logic.

The result is a technology-independent engineering requirement suitable for subsequent runtime analysis.

4.6 Runtime Requirement Identification

Not every operational requirement requires runtime evaluation.

The methodology therefore distinguishes between:

- organisational governance requirements;
- process requirements;
- documentation requirements;
- runtime execution requirements.

Only requirements that influence the legitimacy of execution immediately before consequence formation are carried forward into Runtime Context definition.

This distinction ensures that the Runtime Authority evaluates only those conditions that are relevant at the point of execution.

4.7 Runtime Context Definition

Each runtime requirement is mapped to one or more Runtime Context elements.

Runtime Context represents the observable operational state that must be available immediately before execution.

Typical Runtime Context elements include:

- consultation status;
- documentation review status;
- clinical approval status;
- execution destination;
- operational environment;
- workflow state.

Runtime Context provides the structured information required by the Runtime Authority to perform deterministic evaluation.

4.8 Runtime Rule Derivation

Once Runtime Context has been defined, deterministic Runtime Rules are derived.

Each Runtime Rule evaluates a single engineering condition.

The methodology deliberately avoids creating large composite rules.

Instead, independent rules provide several advantages:

- improved traceability;
- simpler verification;
- clearer explainability;
- easier maintenance;
- more precise Authority Receipts.

This approach allows individual rule outcomes to be independently recorded and reconstructed.

4.9 Engineering Outputs

Completion of the source analysis phase produces the engineering artefacts required for subsequent stages of the methodology.

These include:

- structured operational requirements;
- Runtime Context definitions;
- Runtime Rules;
- traceability references;
- engineering rationale supporting each derived rule.

These outputs provide the foundation for the Traceability Matrix, Runtime Verification Specification and Reference Architecture described in subsequent publications.

4.10 Key Engineering Principle

The methodology is founded on a single governing principle:

Narrative operational guidance is not executed directly. It is first transformed into deterministic engineering artefacts that preserve intent while enabling consistent runtime verification.

This principle underpins every subsequent stage of the ASVH engineering lifecycle and explains why the reference implementation is capable of producing deterministic, explainable and traceable runtime decisions.

5. Runtime Requirement Engineering

5.1 Purpose

Runtime Requirements Engineering represents the transition from operational intent to deterministic runtime evaluation.

The preceding chapters established how NHS England guidance is analysed, interpreted and transformed into operational requirements. This chapter defines the engineering process by which those operational requirements are refined into runtime-verifiable requirements capable of supporting independent Runtime Authority evaluation immediately before execution.

Unlike operational requirements, which describe expected organisational behaviour, runtime requirements describe the specific conditions that must be evaluated at execution time to determine whether delegated authority remains legitimately exercisable.

5.2 Runtime Requirements

A runtime requirement is an implementation-independent engineering statement describing a condition that must be evaluated immediately before execution.

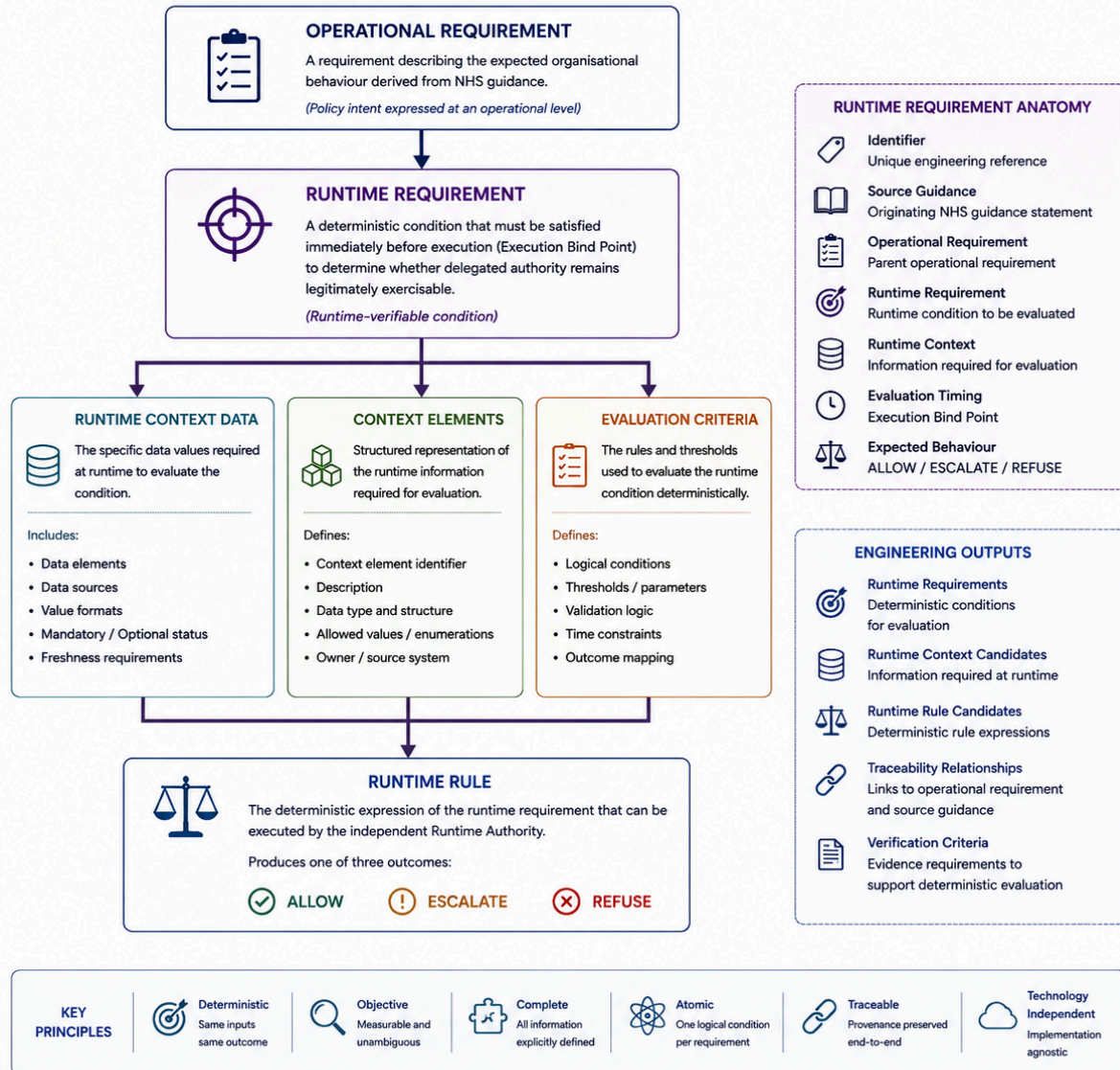
Each runtime requirement shall:

- define an objectively measurable runtime condition;
- identify the Runtime Context information required for evaluation;
- be expressed independently of implementation technology;
- support deterministic evaluation; and
- remain traceable to the originating operational requirement.

Runtime requirements provide the engineering bridge between narrative policy and deterministic Runtime Rules.

Figure 5.1 – Runtime Requirement Structure

Anatomy of a Runtime Requirement and Its Relationship to Deterministic Evaluation



This structure ensures that each runtime requirement is complete, deterministic and independently verifiable at the Execution Bind Point, while remaining fully traceable to its originating guidance and operational intent.

Figure 5.1 – Runtime Requirement Structure

Description

Figure 5.1 illustrates the internal structure of a Runtime Requirement and its relationship to deterministic runtime evaluation. Whereas Figure 4.1 demonstrates how a Runtime Requirement is derived from published NHS England guidance, this figure describes the engineering elements that comprise the Runtime Requirement itself and the information required to support deterministic evaluation by an independent Runtime Authority.

The process begins with an **Operational Requirement** derived from the engineering interpretation of NHS England guidance. This operational requirement expresses the intended organisational behaviour that must be preserved throughout the engineering lifecycle. It is subsequently refined into a **Runtime Requirement**, representing an implementation-independent engineering statement describing a condition that must be evaluated immediately before institutional consequence formation.

Each Runtime Requirement is supported by three complementary engineering components. **Runtime Context Data** defines the information that must be available at execution time, including required data elements, source systems, permitted values and freshness constraints. **Context Elements** provide the structured representation of that information, defining the individual entities, attributes and relationships that comprise the Runtime Context. **Evaluation Criteria** describe the deterministic logic, thresholds and validation conditions that govern runtime evaluation.

Collectively, these engineering components provide the foundation for deriving a deterministic **Runtime Rule**. The Runtime Rule expresses the Runtime Requirement in an executable, implementation-independent form capable of producing one of three deterministic Runtime Decisions: **ALLOW**, **ESCALATE** or **REFUSE**. By separating the Runtime Requirement from the Runtime Rule, the methodology preserves a clear distinction between engineering intent and implementation behaviour.

The figure also identifies the standard engineering metadata associated with every Runtime Requirement, including unique identifiers, originating guidance references, parent operational requirements, Runtime Context dependencies, evaluation timing and expected runtime behaviour. These attributes ensure that every Runtime Requirement remains uniquely identifiable, fully traceable and consistently interpreted throughout the engineering lifecycle.

Finally, Figure 5.1 highlights the engineering outputs generated during Runtime Requirements Engineering, including Runtime Requirement definitions, Runtime Context candidates, Runtime Rule candidates, traceability relationships and verification criteria. Together, these outputs provide the engineering foundation for the subsequent Runtime Context Engineering and Runtime Rule Derivation activities described in the following chapters.

Figure 5.1 establishes the Runtime Requirement as the central engineering artefact within the ASVH Engineering Methodology. It represents the point at which narrative operational intent is transformed into a precise, deterministic and implementation-independent engineering specification capable of supporting reproducible runtime verification while maintaining complete traceability to the originating NHS England guidance.

5.3 Engineering Characteristics

Every runtime requirement should possess the following characteristics.

Deterministic

Evaluation produces identical outcomes when presented with identical Runtime Context.

Objective

Conditions are measurable rather than interpretative.

Complete

All information required for evaluation is explicitly defined.

Atomic

Each runtime requirement evaluates one logical runtime condition.

Traceable

Each runtime requirement maintains provenance to its originating NHS guidance statement.

Technology Independent

Requirements describe engineering behaviour rather than implementation.

5.4 Runtime Requirement Structure

Each runtime requirement should define:

Element

Purpose

Identifier	Unique engineering reference
Source Guidance	Originating NHS guidance
Operational Requirement	Parent engineering requirement
Runtime Requirement	Runtime condition
Runtime Context	Required data
Evaluation Timing	Execution Bind Point
Expected Behaviour	Runtime outcome

This standardised structure supports consistency throughout the engineering lifecycle.

5.5 Engineering Outputs

The outputs produced by Runtime Requirements Engineering become the inputs to subsequent engineering activities.

These outputs include:

- Runtime Requirements;
- Runtime Context candidates;
- Runtime Rule candidates;
- engineering traceability relationships; and
- verification criteria supporting deterministic evaluation.

These artefacts are progressively realised within the companion ASVH publications.

Chapter Summary

This chapter has defined the engineering principles governing Runtime Requirements Engineering and established the characteristics required for deterministic runtime evaluation.

By transforming operational requirements into measurable runtime conditions, the methodology creates the engineering foundation upon which Runtime Context definitions and Runtime Rules are subsequently derived.

The following chapter introduces **Runtime Context Engineering**, describing how the information required for runtime evaluation is systematically identified, structured and maintained.

6. Runtime Rule Derivation

6.1 Introduction

Runtime Requirement Engineering establishes the observable operational state required for deterministic runtime verification. Runtime Rule Derivation transforms those observable states into executable engineering controls capable of supporting independent Runtime Authority decisions.

The purpose of this stage is not to define organisational policy or workflow logic. Rather, it is to create a deterministic evaluation model that can consistently determine whether delegated authority remains exercisable immediately before consequence formation.

Each Runtime Rule therefore represents a discrete engineering control derived from documented Runtime Context and evaluated independently by the Runtime Authority.

The resulting Runtime Rules form the operational decision model described in **ASVH-SPEC-001** and implemented within the **ASVH-001 Reference Implementation**.

6.2 Engineering Objective

The objective of Runtime Rule Derivation is to transform Runtime Context into deterministic evaluation logic while preserving:

- traceability to published guidance;
- implementation independence;
- engineering explainability;
- deterministic behaviour;
- independent verification.

The methodology deliberately separates the derivation of Runtime Rules from the software implementation that ultimately evaluates them.

This distinction ensures that the engineering methodology remains valid regardless of implementation technology.

6.3 Rule Derivation Process

Each Runtime Rule is derived using a structured engineering process.

The process begins with a Runtime Requirement that describes an execution-relevant operational condition.

The Runtime Requirement is then associated with the Runtime Context required to determine whether that condition is satisfied immediately before execution.

From this Runtime Context, a deterministic evaluation condition is defined.

Finally, the evaluation condition is expressed as a Runtime Rule capable of producing a reproducible engineering outcome.

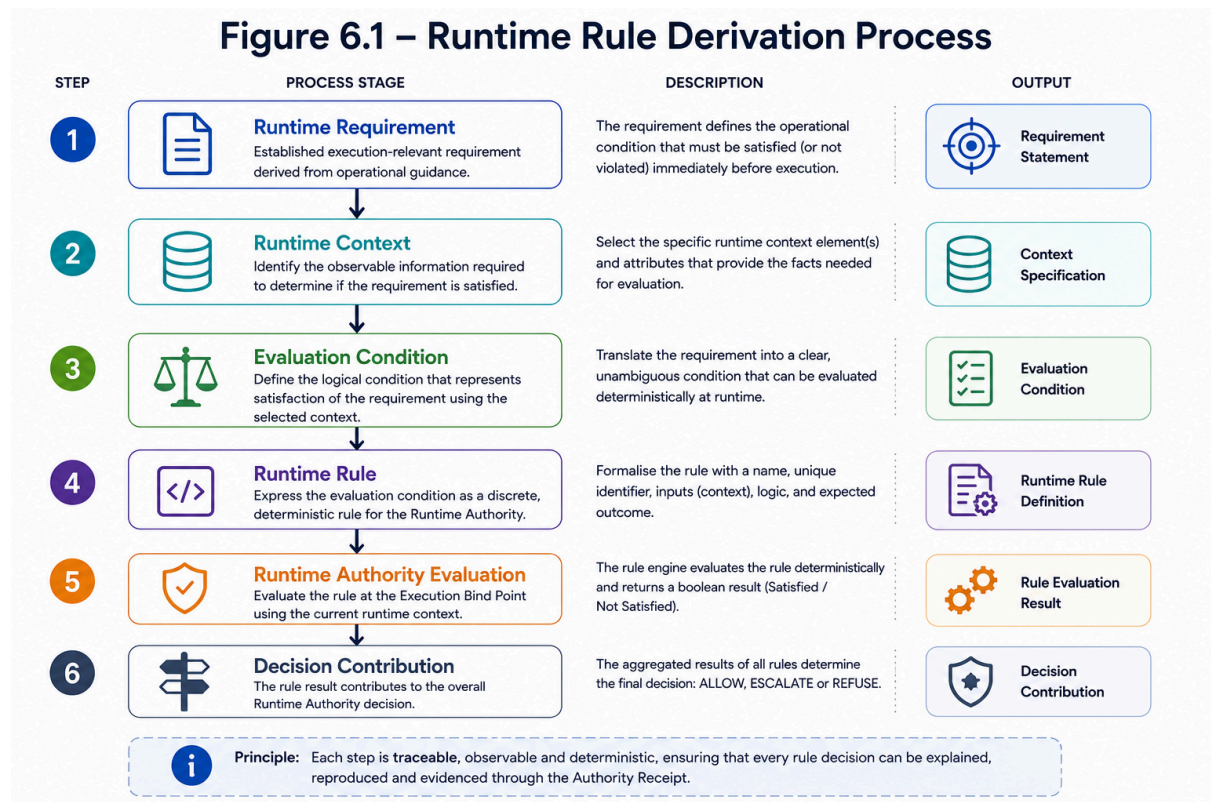


Figure 6.1 – Runtime Rule Derivation Process

Figure 6.1 illustrates the structured engineering process used to derive deterministic Runtime Rules from implementation-independent Runtime Requirements. Whereas the preceding chapters establish how NHS England guidance is transformed into Runtime Requirements and Runtime Context, this figure demonstrates the final engineering transition from documented runtime conditions to executable Runtime Authority behaviour.

The derivation process begins with a Runtime Requirement expressing an execution-relevant operational condition derived from published NHS England guidance. The corresponding Runtime Context is then identified, defining the observable operational information required to determine whether that requirement is satisfied immediately before execution.

Using this Runtime Context, an Evaluation Condition is constructed that expresses the requirement as an objective and deterministic engineering condition. This evaluation condition is subsequently formalised as a Runtime Rule, providing an implementation-independent rule definition suitable for execution by an independent Runtime Authority.

During runtime operation, the Runtime Rule is evaluated against the current Runtime Context at the Execution Bind Point, producing an individual Rule Evaluation Result. This result contributes to the overall Runtime Authority decision by indicating whether the associated operational condition has been satisfied. The Runtime Authority subsequently combines the outcomes of all applicable Runtime Rules to produce one of the deterministic execution decisions defined by the methodology: ALLOW, ESCALATE or REFUSE.

The figure emphasises that each Runtime Rule remains independently derived, independently evaluated and independently traceable throughout the engineering lifecycle. Every stage preserves explicit provenance to the originating Runtime Requirement and supporting Runtime Context, enabling complete reconstruction of the engineering pathway from published guidance through to Runtime Authority decision formation.

Figure 6.1 therefore represents the culmination of the engineering methodology defined by this publication. It demonstrates how narrative operational intent is progressively transformed into deterministic runtime evaluation while preserving implementation independence, engineering explainability, complete traceability and evidential accountability.

This progression ensures that every Runtime Rule remains directly attributable to documented engineering artefacts.

6.4 Characteristics of Runtime Rules

The methodology defines five characteristics that every Runtime Rule should exhibit.

Deterministic

A Runtime Rule shall always produce the same outcome when presented with identical Runtime Context.

Determinism is fundamental to reproducibility and independent verification.

Independent

Each Runtime Rule evaluates a single operational condition.

Rules should not modify the behaviour of other rules or depend upon hidden implementation state.

Independence simplifies testing and improves explainability.

Observable

Every Runtime Rule shall evaluate information that is observable immediately before execution.

Runtime Rules shall not depend upon inferred operational state or speculative future behaviour.

Traceable

Every Runtime Rule maintains documented traceability to:

- published guidance;
- engineering requirements;
- Runtime Context;
- verification scenarios;
- Authority Receipt evidence.

This traceability enables every Runtime Authority decision to be reconstructed through documented engineering artefacts.

Verifiable

Every Runtime Rule shall be associated with one or more verification scenarios demonstrating expected operational behaviour.

Verification scenarios provide objective evidence that the Runtime Rule behaves consistently across representative operational conditions.

6.5 Rule Composition

The methodology intentionally avoids constructing large composite decision algorithms.

Instead, Runtime Authority evaluates a collection of small, independently understandable Runtime Rules.

This approach provides several engineering advantages.

First, individual Runtime Rules are easier to review.

Second, traceability remains straightforward.

Third, Authority Receipts can identify precisely which Runtime Rule influenced a particular decision.

Finally, implementations remain easier to maintain as operational guidance evolves.

The methodology therefore favours composability over complexity.

6.6 Runtime Evaluation

Runtime Rule evaluation occurs immediately before execution at the architectural location identified within the ASVH Reference Architecture as the **Execution Bind Point**.

At this point, the Runtime Authority receives the complete Runtime Context and evaluates each Runtime Rule independently.

Individual Runtime Rule outcomes are then combined to produce a deterministic Runtime Authority decision.

The methodology recognises three decision outcomes.

- **ALLOW** — Runtime verification confirms that delegated authority remains exercisable and execution may proceed.
- **ESCALATE** — Runtime verification identifies conditions requiring further review before execution may continue.
- **REFUSE** — Runtime verification determines that delegated authority is no longer exercisable and execution shall not proceed.

The methodology intentionally defines these outcomes independently of implementation technology.

Different implementations may vary internally while producing equivalent deterministic behaviour.

6.7 Relationship to Companion Publications

This document explains the methodology used to derive Runtime Rules.

It does not define the Runtime Rules themselves.

The engineering outputs generated during this stage are documented within the companion publications.

- **ASVH-SPEC-001** defines the Runtime Rules and their evaluation logic.
- **ASVH-TRACE-001** records traceability between Runtime Rules and published guidance.
- **ASVH-ARCH-001** identifies the architectural location at which Runtime Rule evaluation occurs.
- **ASVH-001** demonstrates one implementation of the Runtime Rule evaluation process.

Together these publications provide a complete engineering representation of deterministic Runtime Authority behaviour.

6.8 Engineering Summary

Runtime Rule Derivation completes the transition from descriptive operational requirements to deterministic runtime evaluation.

By deriving independent Runtime Rules from observable Runtime Context, the methodology creates an engineering model capable of consistent, explainable and reproducible Runtime Authority decisions.

The resulting Runtime Rules provide the foundation for traceability, architectural definition, verification scenario design and executable implementation while preserving a clear separation between engineering methodology and implementation-specific behaviour.

7. Traceability Engineering

7.1 Introduction

A deterministic Runtime Authority is only as trustworthy as the engineering evidence that explains its behaviour. Traceability Engineering provides the disciplined methodology through which every runtime decision can be related to the engineering artefacts, engineering decisions and published NHS England guidance from which it originated.

Achieving this objective requires more than simply maintaining references between documents. It requires a structured traceability methodology that preserves the relationship between published operational guidance, engineering decisions and executable runtime behaviour throughout the complete engineering lifecycle.

Traceability Engineering therefore provides the mechanism through which the provenance of every Runtime Rule, architectural decision and runtime evaluation can be reconstructed.

Rather than treating traceability as documentation produced after implementation, the ASVH methodology considers traceability to be an engineering activity performed throughout the development lifecycle.

7.2 Engineering Objective

The objective of Traceability Engineering is to maintain an unbroken chain of evidence linking every implementation artefact back to its originating engineering decision and, ultimately, to the published NHS England guidance.

This objective supports four engineering outcomes:

- explainability;
- independent review;
- engineering accountability;
- implementation transparency.

By preserving these relationships, the methodology enables every Runtime Authority decision to be understood in terms of documented engineering evidence rather than implementation-specific behaviour.

7.3 Traceability Philosophy

The methodology deliberately avoids many-to-many traceability wherever practical.

Instead, engineering artefacts are related through a sequence of controlled transformations.

Each engineering output becomes the documented input to the next engineering activity.

This approach creates a clear engineering lineage while reducing ambiguity and simplifying independent review.

The resulting engineering chain is illustrated in Figure 7.1.

Figure 7.1 – Traceability Engineering Chain

End-to-end traceability from published guidance to runtime evidence

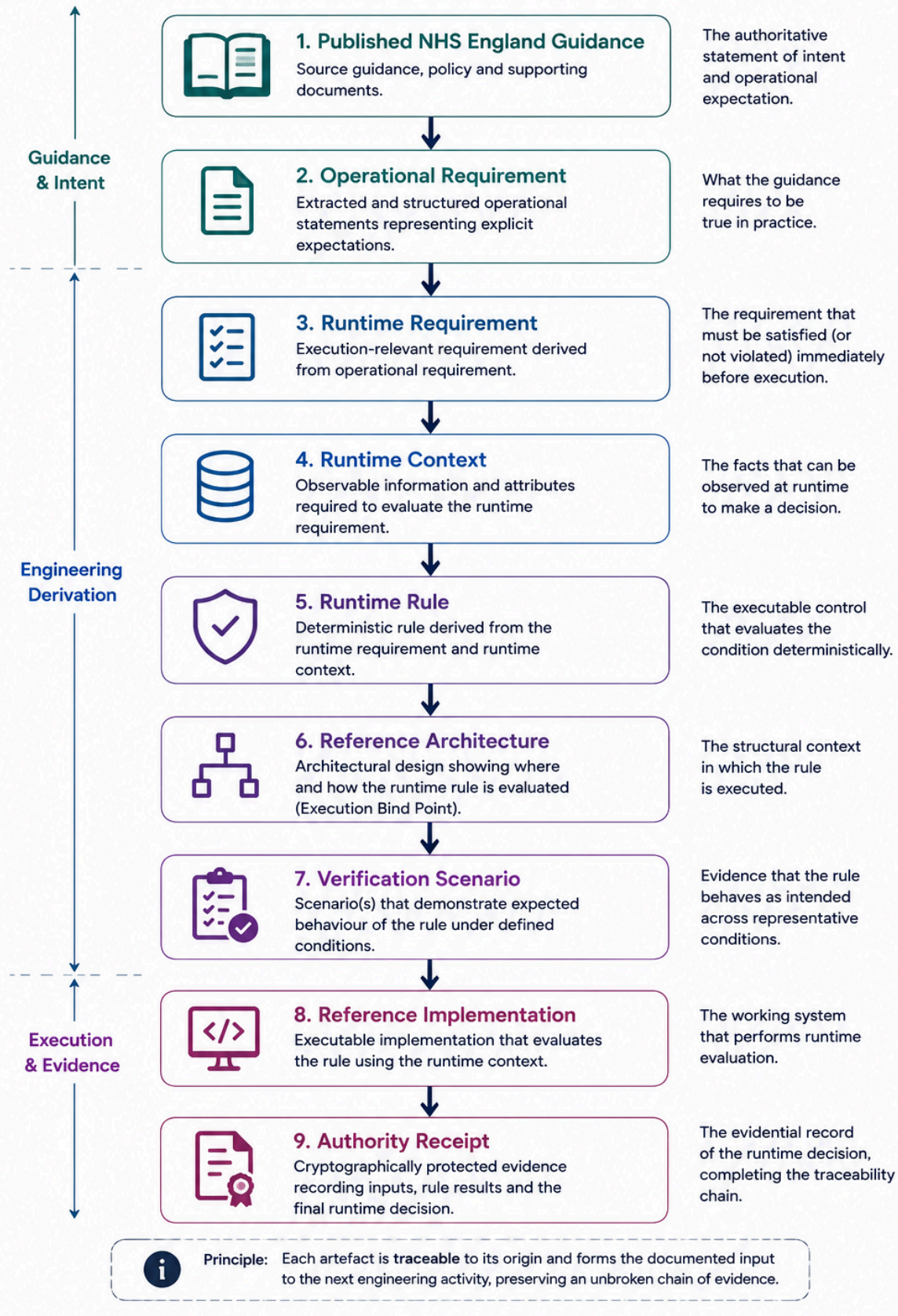


Figure 7.1 – Traceability Engineering Chain

Figure 7.1 illustrates the complete engineering lineage established by the ASVH Engineering Methodology, demonstrating how every engineering artefact is progressively derived from published NHS England guidance while preserving an unbroken chain of engineering provenance. Unlike conventional software traceability models, which typically terminate at implementation, the ASVH methodology extends traceability through runtime execution to the production of evidential Authority Receipts.

The engineering chain begins with **published NHS England guidance**, representing the authoritative source of operational intent. This guidance is systematically transformed into **Operational Requirements**, expressing the expected organisational behaviours that must be preserved throughout the engineering lifecycle. These requirements are subsequently refined into **Runtime Requirements**, defining the execution-relevant conditions that must be evaluated immediately before institutional consequence formation.

From the Runtime Requirements, the methodology derives the required **Runtime Context**, capturing the observable operational information necessary to support deterministic evaluation. This Runtime Context forms the basis for deriving independent **Runtime Rules**, which express the required runtime behaviour in an implementation-independent form suitable for evaluation by an independent Runtime Authority.

The derived Runtime Rules are then realised through the companion engineering publications. The **Reference Architecture** defines where Runtime Rule evaluation occurs within the overall system architecture. **Verification Scenarios** provide objective evidence that each Runtime Rule behaves consistently under representative operational conditions. The **Reference Implementation** demonstrates one implementation of the engineered Runtime Rules without constraining future implementations.

The engineering chain concludes with the generation of an **Authority Receipt**, which records the Runtime Context presented, the Runtime Rules evaluated and the resulting deterministic Runtime Authority decision. The Authority Receipt therefore represents the final evidential link within the engineering lifecycle, enabling runtime decisions to be reconstructed through documented engineering artefacts rather than implementation-specific behaviour.

Figure 7.1 illustrates the fundamental traceability principle governing the ASVH Engineering Methodology: **each engineering artefact is derived from, and remains explicitly linked to, the artefact that preceded it**. This disciplined chain of provenance ensures that deterministic Runtime Authority behaviour can be independently explained, reviewed and verified from published NHS England guidance through to recorded runtime evidence while preserving implementation independence throughout the engineering lifecycle.

7.4 Engineering Relationships

Each engineering activity creates a documented relationship that is preserved throughout the publication series.

These relationships include:

- guidance to operational requirement;
- operational requirement to Runtime Requirement;
- Runtime Requirement to Runtime Context;
- Runtime Context to Runtime Rule;
- Runtime Rule to verification scenario;
- Runtime Rule to architecture;
- Runtime Rule to implementation;
- Runtime Rule to Authority Receipt evidence.

Each relationship is established once and subsequently reused by the companion publications.

This avoids duplication while ensuring consistency across the publication series.

7.5 Traceability Granularity

The methodology intentionally maintains traceability at the level of individual engineering artefacts rather than entire documents.

For example, individual Runtime Rules are linked to the specific Runtime Requirements from which they were derived.

Similarly, individual verification scenarios reference only those Runtime Rules relevant to the operational condition being demonstrated.

This level of granularity supports:

- precise engineering review;
 - simplified maintenance;
 - targeted verification;
 - reproducible Authority Receipts.
-

7.6 Traceability Through Runtime Execution

The ASVH methodology extends engineering traceability beyond design-time artefacts.

During runtime evaluation, the Runtime Authority records the Runtime Rules evaluated, the Runtime Context presented and the resulting deterministic decision.

These observations are preserved within the Authority Receipt.

Consequently, the Authority Receipt represents not only evidence of execution but also the final link within the engineering traceability chain.

The methodology therefore provides traceability from published guidance through to recorded runtime evidence.

7.7 Relationship to the Companion Publications

This document describes the methodology used to construct engineering traceability.

The resulting traceability relationships are documented separately within the ASVH publication series.

- **ASVH-TRACE-001** records the complete Runtime Traceability Matrix.
- **ASVH-SPEC-001** defines the Runtime Rules referenced by the traceability model.
- **ASVH-ARCH-001** identifies the architectural locations associated with Runtime Rule evaluation.
- **ASVH-001** demonstrates runtime traceability through executable evaluation and Authority Receipt generation.

This separation preserves the methodological role of this publication while avoiding duplication of engineering artefacts.

7.8 Engineering Characteristics

The Traceability Engineering methodology has been designed to exhibit five characteristics.

Characteristic	Description	Engineering Benefit
Complete	Every engineering artefact has an identifiable origin.	Every engineering artefact has an identifiable origin.
Continuous	Traceability is maintained throughout the lifecycle.	Traceability is maintained throughout the engineering lifecycle.
Deterministic	Relationships remain stable across implementations.	Traceability relationships remain stable across implementations.

Explainable	Runtime decisions can be reconstructed from engineering artefacts.	Runtime decisions can be reconstructed through documented engineering evidence.
Auditable	Provenance can be verified independently of implementation internals.	Independent reviewers can verify engineering provenance without examining implementation internals.

7.9 Engineering Summary

Traceability Engineering ensures that every engineering artefact produced by the ASVH methodology can be related to the engineering decision from which it originated.

By preserving an unbroken chain from published NHS England guidance to Authority Receipt evidence, the methodology enables Runtime Authority decisions to be independently explained, reviewed and reproduced.

By preserving complete engineering provenance from published guidance through to Authority Receipt evidence, Traceability Engineering provides the evidential foundation upon which deterministic Runtime Authority implementations can be independently reviewed, verified and trusted.

The engineering relationships established during this stage provide the foundation for the Reference Architecture described in the following chapter.

8. Architecture Development

8.1 Introduction

The preceding chapters described the engineering methodology used to transform published NHS England guidance into deterministic Runtime Rules supported by documented Runtime Context and engineering traceability.

Determining what should be evaluated does not, by itself, define how Runtime Authority should be integrated into an operational workflow. A separate engineering activity is therefore required to determine the architectural location at which deterministic runtime verification can be performed while execution remains preventable.

This activity represents the transition from engineering analysis to architectural design.

The objective is not to design a software system or prescribe a technical implementation. Rather, it is to identify the architectural location at which deterministic runtime verification can be performed immediately before clinical consequence formation.

For the ASVH Reference Implementation, this analysis resulted in the identification of the **Execution Bind Point**.

8.2 Engineering Objective

The objective of Architecture Development is to determine the architectural location at which Runtime Authority should evaluate Runtime Context before execution proceeds.

To achieve this objective, the methodology applies four engineering principles.

- Runtime verification shall occur immediately before consequence formation.
- Runtime verification shall remain independent of content generation.
- Runtime verification shall evaluate the complete Runtime Context.
- Runtime verification shall preserve deterministic behaviour irrespective of implementation technology.

These principles ensure that architectural placement is determined by engineering requirements rather than implementation convenience.

8.3 Architectural Analysis

During engineering analysis, the ambient scribing workflow was examined as a sequence of operational responsibilities.

These responsibilities include:

- generation of clinical documentation;
- clinician review;
- clinician approval;
- preparation for execution;
- commitment to the Electronic Patient Record.

Each responsibility contributes to the overall workflow but performs a distinct operational function.

The methodology therefore considers each responsibility independently when determining where Runtime Authority should operate.

The objective is to identify the point at which all relevant Runtime Context has become available while execution remains preventable.

8.4 Identification of the Execution Bind Point

The engineering analysis determined that deterministic runtime verification requires two conditions to exist simultaneously.

First, all Runtime Context required for evaluation must be available.

Second, execution must still be capable of being prevented.

Earlier stages within the workflow do not satisfy the first condition because Runtime Context remains incomplete.

Later stages do not satisfy the second condition because consequence formation has already commenced.

The methodology therefore identifies the architectural boundary immediately preceding execution as the optimal location for Runtime Authority evaluation.

This boundary is designated the **Execution Bind Point**.

The Execution Bind Point represents the final opportunity to determine whether delegated authority remains legitimately exercisable before clinical consequence formation.

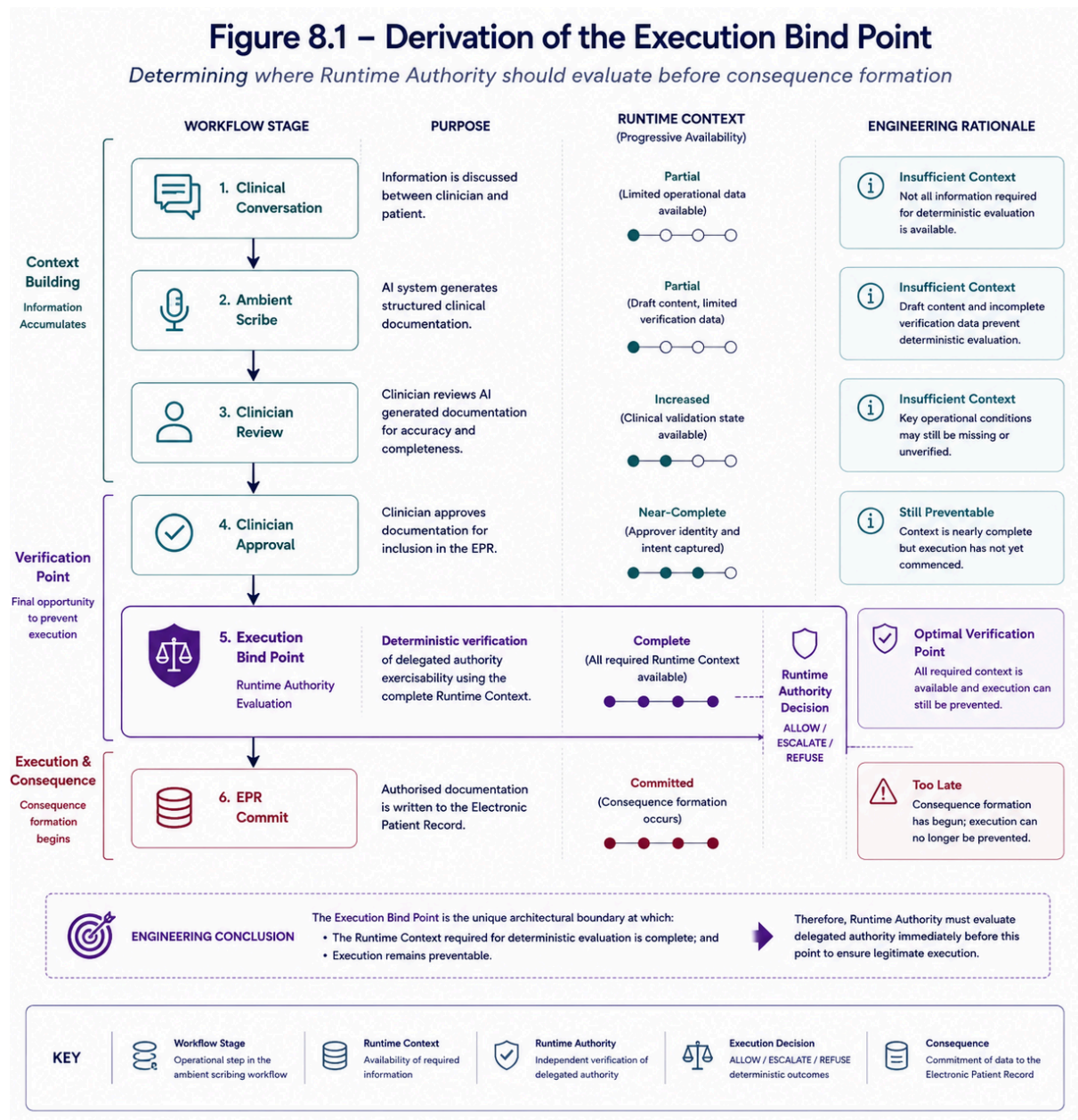


Figure 8.1 – Derivation of the Execution Bind Point

Figure 8.1 illustrates the engineering analysis used by the ASVH Engineering Methodology to identify the **Execution Bind Point** as the appropriate architectural location for deterministic Runtime Authority evaluation. Rather than presenting the completed Reference Architecture, the figure demonstrates the engineering reasoning that determines where runtime verification should occur within the ambient scribing workflow.

The workflow is analysed as a sequence of distinct operational responsibilities beginning with the clinical conversation, followed by AI-assisted documentation generation, clinician review and clinician approval. Throughout these stages, the Runtime Context progressively accumulates as additional operational information becomes available. At each stage prior to

clinician approval, the Runtime Context remains incomplete and therefore insufficient to support deterministic runtime verification.

Following clinician approval, the methodology identifies a unique architectural boundary at which the Runtime Context becomes complete while execution remains preventable. This boundary is designated the **Execution Bind Point**. At this point, the independent Runtime Authority receives the complete Runtime Context and performs deterministic evaluation of the applicable Runtime Rules before any institutional consequence is permitted to occur.

The Runtime Authority returns one of three deterministic execution decisions: **ALLOW**, **ESCALATE** or **REFUSE**. Only an **ALLOW** decision permits commitment of the approved clinical documentation to the Electronic Patient Record (EPR). Where an **ESCALATE** or **REFUSE** decision is returned, execution is prevented in accordance with the evaluated Runtime Context and documented Runtime Rules.

The figure also distinguishes three engineering phases within the workflow. During **Context Building**, operational information progressively accumulates but deterministic evaluation cannot yet be performed. At the **Verification Point**, the Runtime Context is complete and execution remains preventable, making deterministic Runtime Authority evaluation both possible and meaningful. During **Execution and Consequence Formation**, commitment to the Electronic Patient Record has commenced and preventive runtime verification is no longer possible.

Figure 8.1 therefore demonstrates the principal architectural conclusion of the ASVH Engineering Methodology: the Execution Bind Point is the unique location within the operational workflow where all information required for deterministic runtime evaluation is available while execution can still be prevented. This engineering conclusion provides the methodological basis for the Reference Architecture defined separately in **ASVH-ARCH-001**, while preserving a clear distinction between architectural derivation and architectural specification.

8.5 Separation of Architectural Responsibilities

Identification of the Execution Bind Point also establishes a clear separation between operational responsibilities within the workflow.

The ambient scribing product remains responsible for generating clinical documentation.

The responsible clinician remains accountable for reviewing and approving that documentation.

The Runtime Authority evaluates whether delegated authority remains exercisable immediately before execution.

The Electronic Patient Record remains responsible for recording the authorised clinical consequence.

This separation ensures that Runtime Authority neither replaces nor duplicates the responsibilities of adjacent architectural components.

Instead, it provides an independent verification function immediately before execution.

8.6 Architectural Independence

The methodology intentionally separates Runtime Authority from implementation-specific workflow components.

Runtime Authority is not embedded within:

- the ambient scribing application;
- the Electronic Patient Record;
- workflow orchestration;
- user interface logic.

Instead, Runtime Authority is positioned as an independent architectural capability responsible solely for deterministic runtime verification.

This separation preserves implementation independence and allows equivalent Runtime Authority behaviour to be achieved across multiple software architectures.

8.7 Relationship to Companion Publications

This document explains the engineering methodology used to derive the Reference Architecture.

The resulting architecture is documented separately within the ASVH publication series.

- **ASVH-ARCH-001** defines the Reference Architecture.
- **ASVH-SPEC-001** defines the Runtime Rules evaluated at the Execution Bind Point.
- **ASVH-TRACE-001** records the engineering relationships supporting the architectural decisions.
- **ASVH-001** demonstrates one implementation of the Reference Architecture.

The methodology therefore explains **how** the architecture was derived rather than describing the architecture itself.

8.8 Engineering Characteristics

The Architecture Development methodology exhibits five engineering characteristics.

Characteristic	Engineering Benefit
Independent	Runtime verification remains separate from content generation and consequence execution.
Deterministic	Runtime evaluation occurs at a consistent architectural location.
Implementation Independent	Architectural principles remain valid across multiple technologies.
Observable	Complete Runtime Context is available before evaluation.
Preventative	Execution can still be prevented before consequence formation.

8.9 Engineering Summary

Architecture Development completes the transition from engineering analysis to architectural design by identifying the location at which deterministic runtime verification should occur.

The methodology concludes that Runtime Authority is most effectively positioned at the **Execution Bind Point**, where complete Runtime Context is available while execution remains preventable.

By identifying the Execution Bind Point as the unique architectural location at which complete Runtime Context is available while execution remains preventable, the methodology establishes the engineering basis for implementation-independent Runtime Authority architectures. This conclusion links the engineering methodology defined within this publication to the Reference Architecture described in ASVH-ARCH-001 while preserving a clear separation between methodology and architectural specification.

This architectural separation preserves implementation independence, supports deterministic evaluation and provides the structural foundation for the verification strategy described in the following chapter.

9. Verification Strategy

9.1 Introduction

The preceding chapters define how published NHS England guidance is systematically transformed into deterministic engineering artefacts, including Runtime Requirements, Runtime Context, Runtime Rules, engineering traceability and architectural separation. Before these artefacts can be realised within a reference implementation, the methodology requires objective verification that they produce consistent, explainable and reproducible Runtime Authority behaviour under representative operational conditions.

Verification Strategy therefore provides the structured process through which the derived Runtime Rules are evaluated prior to implementation release.

The objective is not to validate clinical policy or organisational governance. Rather, it is to demonstrate that the engineering artefacts produced by the methodology exhibit deterministic, explainable and repeatable behaviour when presented with defined Runtime Context.

9.2 Engineering Objective

The Verification Strategy has four objectives.

- Demonstrate deterministic Runtime Rule behaviour.
- Validate engineering traceability through representative scenarios.
- Confirm that Runtime Authority decisions are reproducible.
- Provide objective evidence supporting the reference implementation.

Verification therefore acts as the engineering assurance activity linking design-time artefacts with executable behaviour.

9.3 Verification Philosophy

Verification within the ASVH methodology is scenario-driven.

Rather than testing individual software functions in isolation, the methodology evaluates Runtime Authority behaviour using representative operational scenarios.

Each scenario defines:

- the Runtime Context presented to the Runtime Authority;
- the Runtime Rules expected to be evaluated;

- the anticipated deterministic outcome;
- the Authority Receipt evidence expected to be produced.

This approach allows engineering decisions to be validated independently of implementation technology.

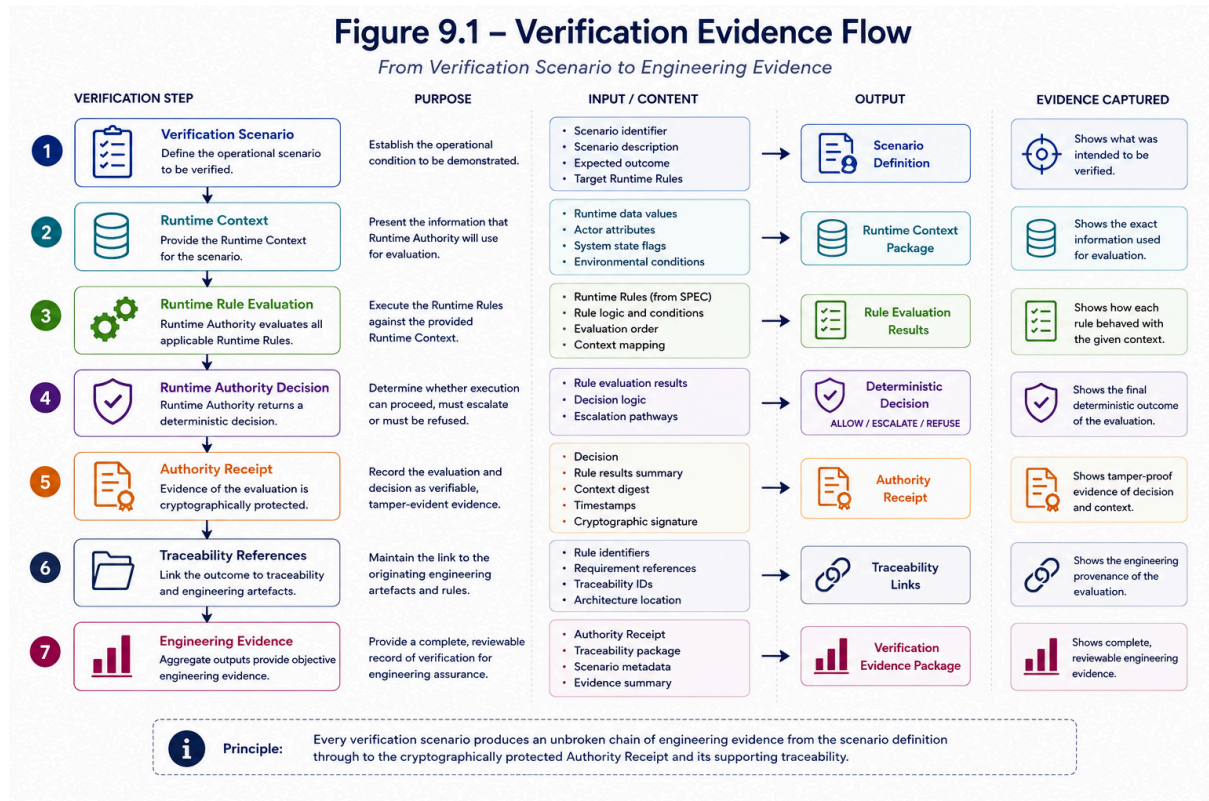


Figure 9.1 – Verification Evidence Flow

Figure 9.1 illustrates the engineering verification process defined by the ASVH Engineering Methodology, demonstrating how representative verification scenarios are transformed into objective engineering evidence supporting deterministic Runtime Authority behaviour. Rather than depicting software test execution, the figure represents the structured engineering process through which the methodology validates the behaviour of the derived Runtime Rules and their associated Runtime Context.

The process begins with the definition of a **Verification Scenario**, which specifies a representative operational condition to be demonstrated. Each scenario identifies the expected Runtime Context, the Runtime Rules that should be evaluated and the anticipated deterministic Runtime Authority outcome. This scenario provides the controlled engineering conditions under which verification is performed.

The defined **Runtime Context** is then assembled and presented to the Runtime Authority. This Runtime Context represents the complete operational information required for deterministic evaluation and forms the authoritative input to the verification process. Using this Runtime Context, the Runtime Authority independently evaluates each applicable

Runtime Rule, producing individual rule evaluation results that collectively contribute to the overall Runtime Authority decision.

The Runtime Authority subsequently returns a deterministic execution decision of **ALLOW**, **ESCALATE** or **REFUSE**. This decision is recorded within an **Authority Receipt**, together with the Runtime Context presented, the Runtime Rules evaluated, timestamps, decision metadata and other supporting engineering evidence required to demonstrate reproducibility and evidential accountability.

Following execution, the methodology establishes explicit **Traceability References** linking the verification outcome to the originating Runtime Requirements, Runtime Rules, architectural decisions and engineering artefacts. These traceability relationships ensure that every verification result remains fully attributable to the engineering decisions from which it originated.

The process concludes with the creation of a consolidated **Verification Evidence Package**, providing objective engineering evidence that the methodology has produced deterministic, explainable and reproducible Runtime Authority behaviour. This evidence package supports independent engineering review without requiring knowledge of implementation-specific software internals.

Figure 9.1 therefore demonstrates the final engineering activity defined by the ASVH Engineering Methodology. It illustrates that verification is not solely concerned with confirming software correctness, but with providing objective evidence that the engineering artefacts derived from published NHS England guidance consistently produce deterministic Runtime Authority decisions. Together with the Authority Receipt and supporting traceability, the verification process completes the unbroken engineering chain established throughout this publication, linking published guidance to independently reviewable runtime evidence.

9.4 Verification Scenario Design

Verification scenarios are intentionally selected to demonstrate both expected and exceptional operational conditions.

The methodology recommends that a representative verification suite include:

- expected operational behaviour;
- incomplete workflow states;
- invalid Runtime Context;
- execution pathway failures;
- unsupported execution requests;
- operational environment failures.

Collectively, these scenarios demonstrate that Runtime Authority decisions remain consistent across a representative range of operational conditions.

9.5 Deterministic Evaluation

Each verification scenario is executed using the Runtime Context defined for that scenario.

Runtime Authority evaluates the corresponding Runtime Rules and produces a deterministic decision.

The methodology requires that repeated execution using identical Runtime Context shall produce identical Runtime Authority outcomes.

Any variation indicates either incomplete Runtime Context, ambiguous Runtime Rules or implementation inconsistency, each of which requires engineering review.

9.6 Verification Evidence

Successful verification produces objective engineering evidence including:

- Runtime Context presented for evaluation;
- Runtime Rules evaluated;
- individual Runtime Rule evaluation results;
- deterministic Runtime Authority decision;
- engineering traceability references;

These outputs provide confidence that the reference implementation accurately represents the engineering methodology described throughout this publication.

9.7 Relationship to Companion Publications

This document describes the methodology used to verify engineering artefacts.

Individual verification scenarios and their execution are documented within the companion publications and reference implementation.

- **ASVH-001** demonstrates scenario execution.
 - **ASVH-SPEC-001** defines the Runtime Rules being evaluated.
 - **ASVH-TRACE-001** records traceability relationships supporting each scenario.
-

9.8 Engineering Summary

Verification Strategy concludes the ASVH Engineering Methodology by providing objective evidence that the engineering artefacts derived from published NHS England guidance consistently produce deterministic Runtime Authority behaviour. Through representative verification scenarios, complete engineering traceability and evidential Authority Receipts, the methodology demonstrates that runtime decisions remain reproducible, explainable and independently reviewable while preserving implementation independence.

By validating Runtime Rules through deterministic scenarios, the methodology demonstrates that Runtime Authority decisions are reproducible, explainable and supported by documented engineering evidence.

10. Reference Implementation

10.1 Introduction

The ASVH Reference Implementation represents the practical realisation of the engineering methodology described throughout this publication. While the preceding chapters define how deterministic engineering artefacts are systematically derived from published NHS England guidance, the Reference Implementation demonstrates that those artefacts can be realised within an executable Runtime Authority while preserving implementation independence, deterministic behaviour and complete engineering traceability.

Its purpose is not to define the methodology itself, but to demonstrate one executable implementation of the engineering artefacts derived through the preceding stages of analysis, decomposition, architectural design and verification.

The Reference Implementation therefore serves as an executable representation of the methodology rather than the methodology itself.

10.2 Engineering Objective

The objective of the Reference Implementation is to demonstrate that the engineering methodology can be realised as a functioning Runtime Authority capable of:

- evaluating Runtime Context;
- executing deterministic Runtime Rules;
- producing Runtime Authority decisions;
- generating Authority Receipts;
- preserving engineering traceability.

The implementation exists to illustrate the methodology rather than prescribe a production architecture.

10.3 Implementation Independence

The methodology intentionally avoids prescribing software architecture, programming language or deployment model.

Alternative implementations may differ in their internal design while remaining faithful to the engineering methodology described within this publication.

Conformance is therefore determined by adherence to the engineering principles rather than implementation technology.

10.4 Demonstration Purpose

The ASVH Reference Implementation has three purposes.

First, it demonstrates that the methodology is executable.

Second, it provides a practical environment for engineering review and discussion.

Third, it illustrates how Runtime Authority produces deterministic decisions supported by traceable evidence.

The implementation should therefore be regarded as an engineering demonstration rather than a production clinical system.

The implementation is intended to demonstrate engineering feasibility and support technical review. It should not be interpreted as prescribing production deployment architecture or clinical operating procedures.

10.5 Relationship to Companion Publications

This document explains the engineering methodology leading to the Reference Implementation.

Implementation details are documented separately within the ASVH publication series.

- **ASVH-IMPL-001** describes the implementation architecture.
 - **ASVH-ARCH-001** defines the Reference Architecture.
 - **ASVH-001** provides the executable reference implementation.
-

10.6 Engineering Summary

The Reference Implementation demonstrates that the engineering methodology described throughout this publication can be realised as an executable Runtime Authority while preserving determinism, traceability and implementation independence.

It represents one implementation of the methodology rather than the methodology itself.

11. Conclusions

11.1 Summary

This publication has described the engineering methodology used to develop the NHS Ambient Scribing Runtime Verification Harness.

Beginning with published NHS England guidance, the methodology demonstrates how narrative operational expectations can be systematically transformed into deterministic Runtime Requirements, Runtime Context, Runtime Rules, engineering traceability, architectural separation and executable Runtime Authority behaviour.

Throughout the methodology, emphasis has been placed on preserving traceability, implementation independence and deterministic evaluation.

11.2 Engineering Contribution

The principal contribution of the methodology is the definition of a repeatable engineering process for deriving deterministic runtime verification from published operational guidance.

Rather than introducing new governance obligations or redefining clinical policy, the methodology provides a structured means of expressing existing operational expectations as independently verifiable engineering artefacts.

The resulting Reference Implementation demonstrates one practical application of this methodology.

11.3 Relationship to the ASVH Publication Series

This publication serves as the methodological foundation for the ASVH publication series.

Companion publications provide detailed descriptions of the engineering artefacts produced during each stage of the methodology, including traceability, specification, architecture and implementation.

Together, these publications form a coherent engineering reference describing both the derivation and execution of deterministic Runtime Authority.

11.4 Future Development

The methodology described within this publication has been developed specifically for the ASVH Reference Implementation.

Future work may investigate its application to additional domains or operational guidance. Such work would require independent engineering analysis and should not be inferred from the present publication.

11.5 Closing Statement

The ASVH Engineering Methodology demonstrates that published NHS England guidance can be transformed through disciplined systems engineering into deterministic Runtime Authority behaviour supported by complete engineering traceability, architectural independence and objective runtime evidence. By defining a repeatable methodology rather than a specific implementation, the publication establishes an implementation-independent engineering foundation upon which Runtime Authority systems can be consistently designed, implemented, verified and independently reviewed.

Appendix A – End-to-End Engineering Example

A.1 Purpose

This appendix demonstrates the engineering methodology described throughout this publication using a complete execution from the ASVH Reference Implementation.

Unlike the preceding chapters, which describe the methodology in abstract engineering terms, this appendix follows a single execution path through the complete engineering lifecycle. Beginning with published operational guidance, it demonstrates how narrative operational expectations are transformed into deterministic Runtime Rules, evaluated by the Runtime Authority and recorded within an Authority Receipt immediately before consequence formation.

The purpose of this appendix is to illustrate the application of the methodology. It does not replace or redefine the companion publications, which remain the authoritative source for their respective engineering artefacts.

This appendix provides a practical demonstration of the complete engineering methodology described throughout this publication. Rather than introducing additional engineering concepts, it illustrates how the individual engineering artefacts produced by each stage of the methodology interact during a representative Runtime Authority evaluation. The appendix should therefore be read as a practical companion to the main chapters rather than as additional normative guidance.

A.2 Case Study Overview

The case study follows a representative ambient scribing workflow in which a clinical encounter has been documented and is approaching commitment to the Electronic Patient Record.

Prior to execution, the Runtime Authority evaluates the observable Runtime Context derived from the workflow and determines whether delegated authority remains legitimately exercisable. Depending upon the Runtime Context presented, execution may be allowed, escalated for further review or refused.

This execution path demonstrates the interaction between the engineering artefacts produced throughout the ASVH publication series.

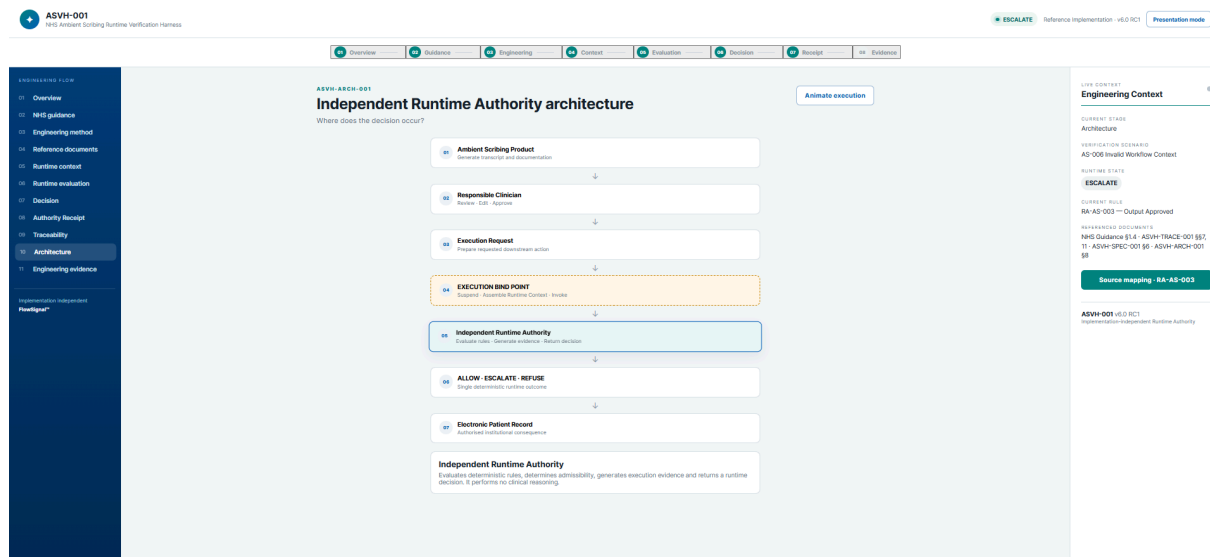


Figure A.1 – Engineering overview of the ASVH Runtime Verification process

A.3 Runtime Context

Immediately prior to execution, the Runtime Authority receives the observable Runtime Context required to evaluate the applicable Runtime Rules.

Runtime Context represents the operational state immediately before consequence formation rather than the complete workflow history. Only those observations required to determine whether delegated authority remains exercisable are presented.

The Runtime Context shown in this example represents the information available immediately before commitment to the Electronic Patient Record.

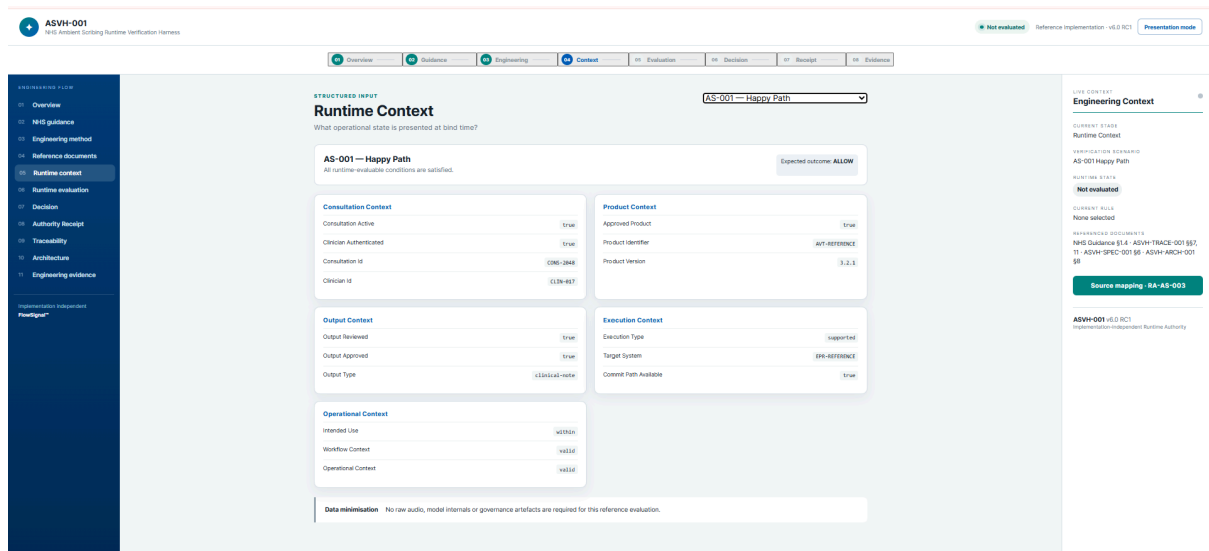


Figure A.2 – Runtime Context presented to the Runtime Authority

A.4 Runtime Rule Evaluation

Using the Runtime Context presented in the previous stage, the Runtime Authority evaluates the deterministic Runtime Rules defined within **ASVH-SPEC-001**. Each Runtime Rule represents an independent engineering control derived through the methodology described in this publication and evaluates a single execution-relevant condition.

Evaluation takes place at the **Execution Bind Point**, where the complete Runtime Context is available and execution remains preventable. The Runtime Authority assesses each Runtime Rule independently before aggregating the individual outcomes into a single deterministic execution decision.

In this case study, all Runtime Rules evaluate successfully, confirming that the operational conditions required for execution remain satisfied. The Runtime Authority therefore authorises progression to the next stage of the workflow and initiates generation of the corresponding Authority Receipt.

The evaluation screen also demonstrates the preservation of engineering traceability throughout runtime execution. Each Runtime Rule is identified individually, its evaluation outcome is displayed, and direct references are maintained to the originating engineering artefacts. This allows every execution decision to be reconstructed using documented engineering evidence rather than implementation-specific behaviour.

Unlike traditional workflow engines, the Runtime Authority does not execute the requested consequence directly. Instead, it independently determines whether delegated authority remains legitimately exercisable immediately before consequence formation.

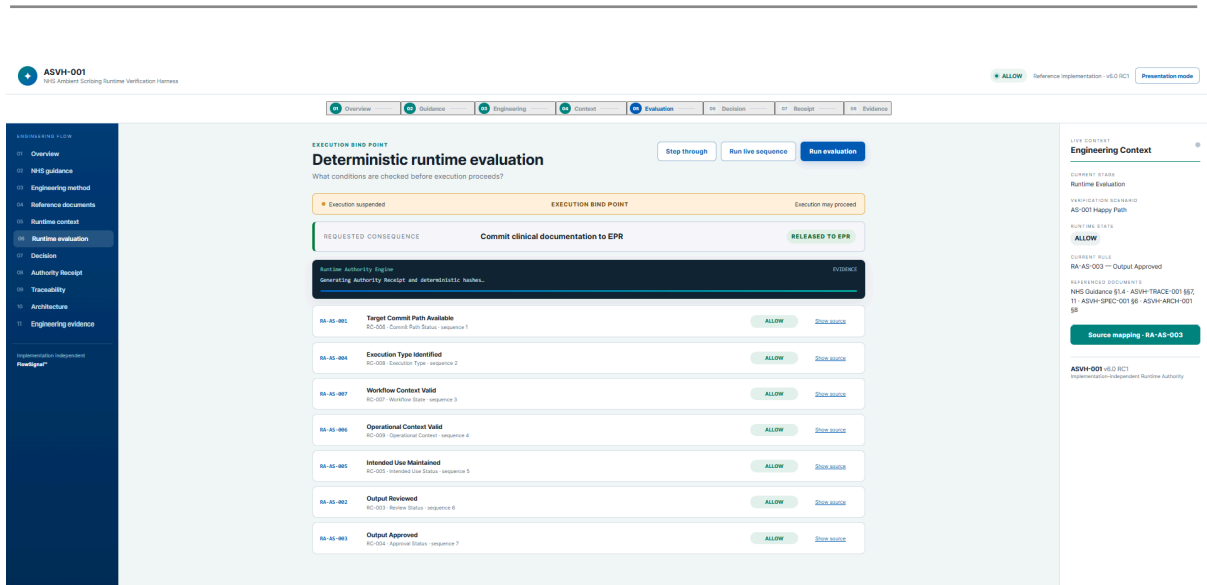


Figure A.3 – Deterministic Runtime Rule evaluation at the Execution Bind Point

A.5 Execution Bind Point

Completion of Runtime Rule evaluation brings the workflow to the Execution Bind Point.

At this architectural boundary all Runtime Context has become available while execution remains preventable.

The Runtime Authority therefore determines whether execution should proceed before institutional consequence formation occurs.

This architectural position is fundamental to the ASVH methodology and provides the independent runtime verification capability described throughout this publication.

The Execution Bind Point represents the final engineering control established by the methodology before execution. It separates operational preparation from execution, ensuring that every consequence is preceded by an independent, deterministic assessment of whether delegated authority remains legitimately exercisable. This architectural separation preserves implementation independence while providing a consistent location for Runtime Authority evaluation across implementations.

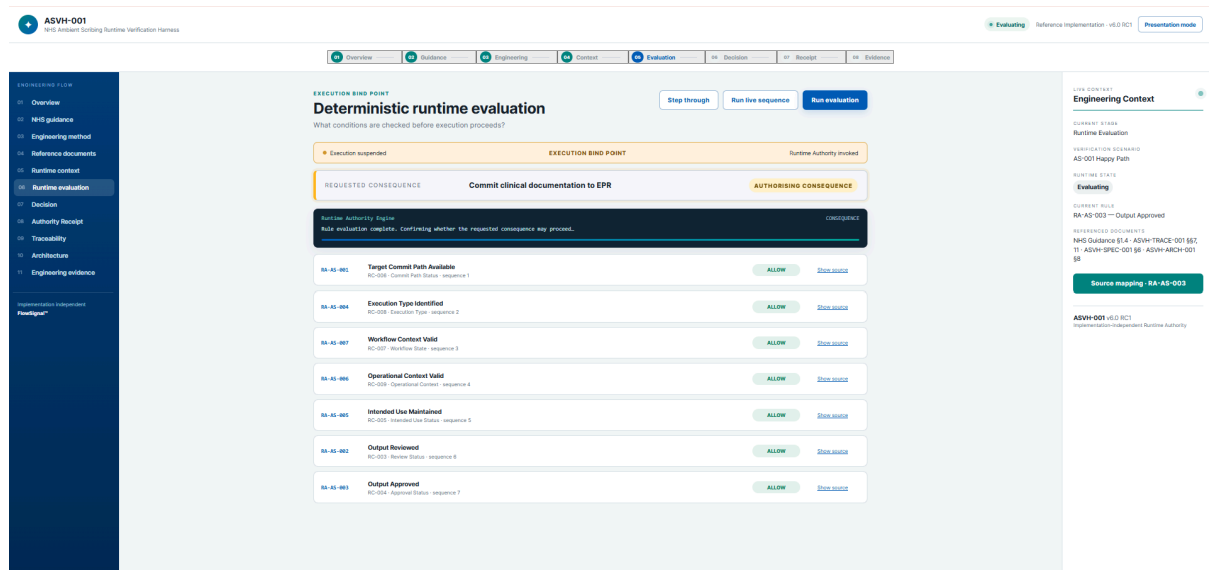


Figure A.4 – Runtime Authority decision

A.6 Runtime Authority Decision

Following evaluation of the applicable Runtime Rules, the Runtime Authority produces a deterministic execution decision.

Within the ASVH Reference Implementation the possible outcomes are:

- ALLOW
- ESCALATE
- REFUSE

Each decision represents the outcome of deterministic Runtime Rule evaluation rather than discretionary judgement.

The Runtime Authority decision represents the aggregation of all Runtime Rule evaluations performed during the preceding stage. Individual Runtime Rules contribute engineering evidence to the decision, but the Runtime Authority remains responsible for determining whether delegated authority continues to be legitimately exercisable immediately before consequence formation. The resulting decision therefore represents the deterministic outcome of the engineering methodology rather than a discretionary judgement by the implementation.

The Runtime Authority does not determine whether the requested action is desirable; it determines whether delegated authority remains legitimately exercisable immediately before execution.

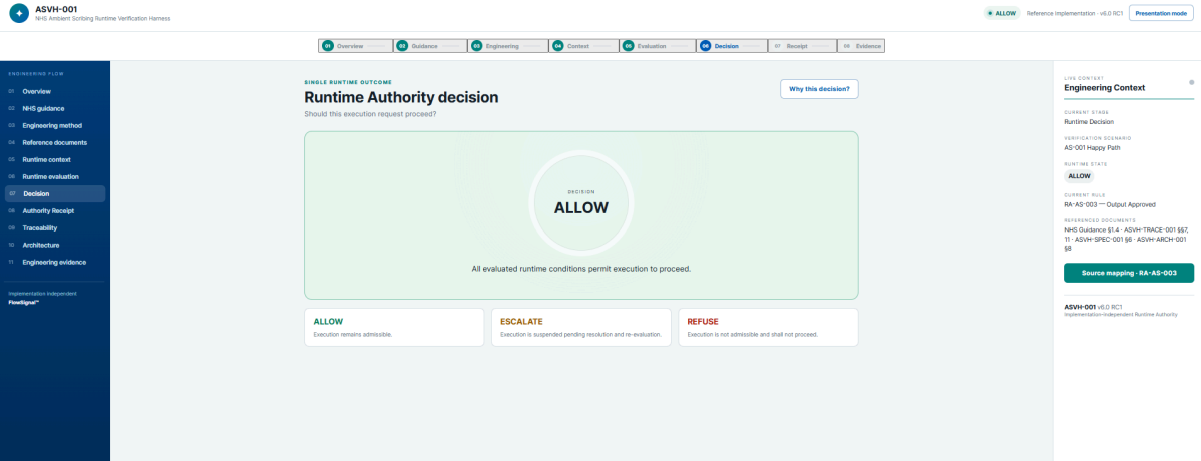


Figure A.5 – Runtime Authority decision

A.7 Authority Receipt

Following determination of the Runtime Authority decision, an Authority Receipt is generated.

The Authority Receipt records the Runtime Context presented, the Runtime Rules evaluated, the Runtime Authority decision and the supporting engineering evidence required to reconstruct the execution decision.

Rather than acting as an audit log, the Authority Receipt provides a structured evidential record demonstrating why delegated authority remained, or ceased to remain, legitimately exercisable immediately before consequence formation.

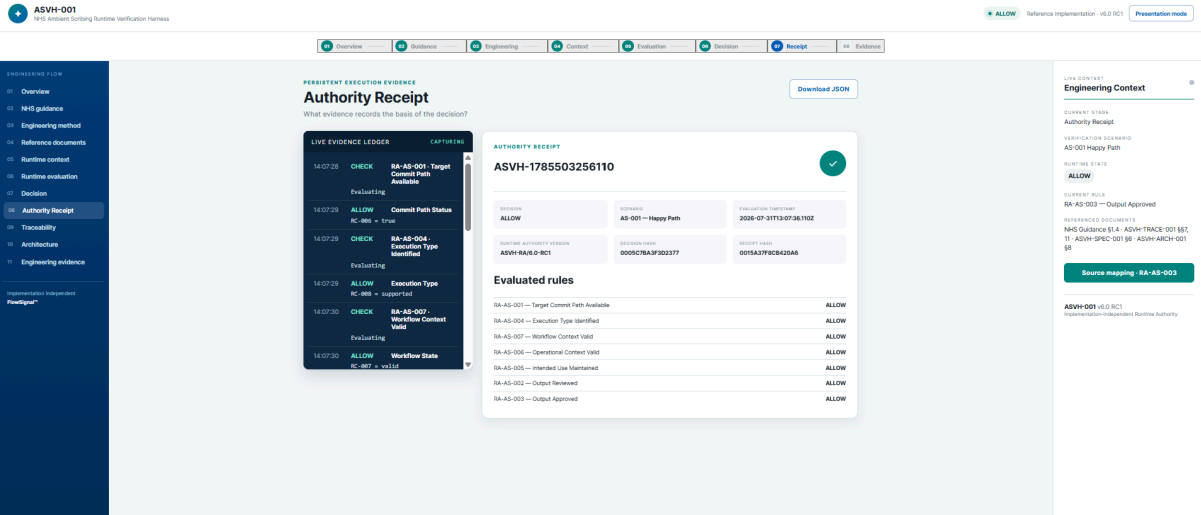


Figure A.6 – Authority Receipt produced by the Runtime Authority

A.8 Traceability

The execution demonstrated within this appendix remains traceable throughout the ASVH publication series.

The Runtime Context presented to the Runtime Authority originates from the Runtime Requirement Engineering process described within this publication.

The Runtime Rules evaluated during execution are specified within ASVH-SPEC-001.

Engineering relationships are maintained within ASVH-TRACE-001.

The architectural location of the Runtime Authority is defined within ASVH-ARCH-001.

The executable behaviour is demonstrated by the ASVH Reference Implementation.

Collectively, these publications provide a complete engineering record from published operational guidance to deterministic runtime verification.

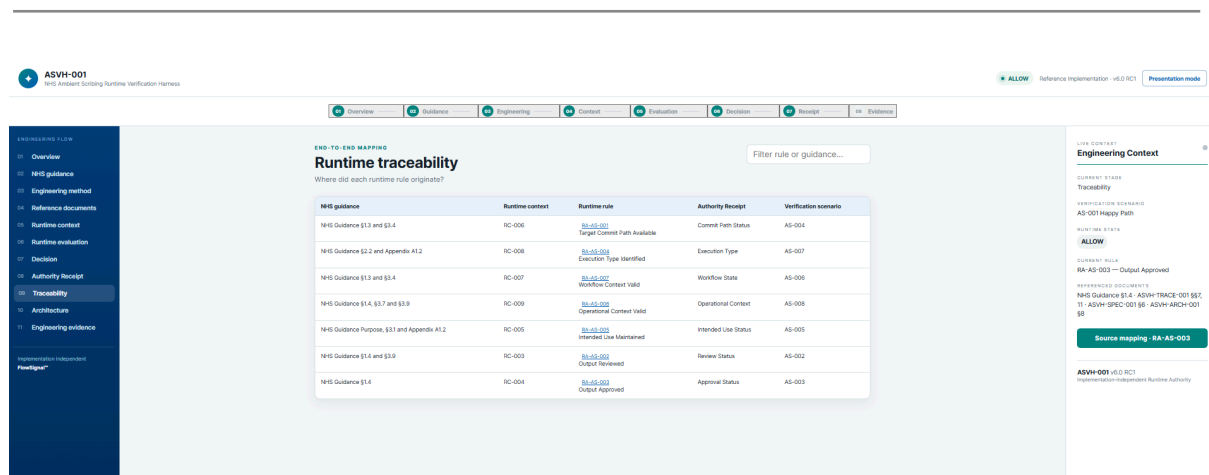


Figure A. 7– Engineering traceability across the ASVH publication series

This provides a satisfying visual conclusion to the case study and reinforces that the methodology has been fully realised in an executable reference implementation.

A.9 Closing Observation

This engineering case study demonstrates the complete transformation described throughout this publication. Beginning with published NHS England guidance, the methodology derives Runtime Requirements, Runtime Context, Runtime Rules, engineering traceability and architectural placement before demonstrating deterministic runtime evaluation through the ASVH Reference Implementation.

This worked engineering example demonstrates that the ASVH Engineering Methodology is not merely a conceptual framework but a complete systems engineering process capable of producing deterministic Runtime Authority behaviour supported by traceable engineering artefacts and objective runtime evidence. By following a single execution from published NHS England guidance through to Authority Receipt generation, the appendix illustrates how the companion publications operate together as a unified engineering framework, providing complete provenance from operational intent to institutional consequence formation.

Appendix B – Related Work and References

B.1 Purpose

The ASVH Engineering Methodology has been developed through the application of established systems engineering principles to the specific challenge of deriving deterministic Runtime Authority behaviour from published operational guidance.

This appendix identifies the principal engineering disciplines, standards and publications that informed the development of the methodology. It is not intended to provide a comprehensive literature review or normative specification. Rather, it acknowledges the engineering foundations upon which the methodology has been developed and identifies the principal contribution made by the ASVH engineering framework.

The methodology does not claim originality for established engineering concepts such as requirements engineering, systems architecture, engineering traceability or verification. Its contribution lies in the structured application of these disciplines to transform published NHS England operational guidance into implementation-independent Runtime Requirements, Runtime Context, Runtime Rules and deterministic Runtime Authority behaviour suitable for execution immediately before execution.

B.2 Engineering Influences

The ASVH Engineering Methodology draws upon several established engineering disciplines.

Engineering Discipline	Contribution to the Methodology
Systems Engineering	Structured engineering lifecycle and disciplined decomposition of complex operational problems.

Requirements Engineering	Transformation of narrative operational guidance into measurable engineering requirements.
Runtime Verification	Evaluation of operational conditions immediately before execution using observable runtime information.
Traceability Engineering	Maintenance of complete engineering provenance throughout the lifecycle.
Software Architecture	Separation of engineering responsibilities and implementation-independent architectural reasoning.
Verification and Validation	Objective engineering evidence supporting deterministic behaviour.
Clinical Safety Engineering	Preservation of clinical accountability and operational safety principles within healthcare environments.

The methodology combines these established engineering disciplines into a coherent process specifically designed to support deterministic Runtime Authority implementations.

B.3 NHS England Guidance

The engineering methodology has been developed using the published NHS England guidance for AI-enabled ambient scribing products in health and care settings as its primary operational source.

Rather than interpreting this guidance as software requirements, the methodology systematically decomposes published operational expectations into engineering artefacts suitable for deterministic runtime verification. Throughout this process, the intent of the originating guidance is preserved while implementation-specific assumptions are deliberately avoided.

The methodology therefore demonstrates one structured approach to engineering implementation while remaining independent of any particular ambient scribing supplier, Electronic Patient Record platform or software technology.

B.4 Relationship to Existing Engineering Practice

The ASVH Engineering Methodology applies established engineering practices in a novel sequence.

Traditional engineering approaches commonly describe:

- requirements engineering;
- software architecture;
- implementation;
- verification.

The ASVH methodology extends this sequence by introducing:

- Runtime Requirement Engineering;
- Runtime Context Engineering;
- deterministic Runtime Rule Derivation;
- Execution Bind Point identification;
- Authority Receipt evidence generation.

These additional engineering activities establish a structured pathway from published operational guidance through to deterministic runtime verification while preserving complete engineering traceability.

B.5 Principal Engineering Contribution

The principal contribution of the ASVH Engineering Methodology is not the introduction of new governance principles or clinical policy.

Instead, it provides a structured engineering methodology that enables published operational guidance to be transformed into deterministic Runtime Authority behaviour while preserving:

- implementation independence;
- engineering traceability;
- deterministic evaluation;
- architectural separation;
- evidential accountability.

The resulting methodology provides a repeatable engineering process capable of supporting independent Runtime Authority implementations across different healthcare technologies and deployment environments.

B.6 Companion Publications

The engineering methodology described within this publication forms part of the wider ASVH engineering framework.

Its outputs are progressively realised through the companion publications:

Publication	Primary Engineering Responsibility
ASVH-METH-001	Engineering methodology and requirements derivation.
ASVH-TRACE-001	Engineering traceability and provenance.
ASVH-SPEC-001	Runtime Rules and Runtime Authority specification.
ASVH-ARCH-001	Implementation-independent Runtime Architecture.
ASVH-001	Reference implementation and engineering verification.

Together, these publications provide a complete engineering framework describing both the derivation and the execution of deterministic Runtime Authority behaviour.

B.7 References

The following publications informed the development of this methodology:

- NHS England. *Guidance on the Use of AI-enabled Ambient Scribing Products in Health and Care Settings*.
- International Council on Systems Engineering (INCOSE). *Systems Engineering Handbook*.
- ISO/IEC/IEEE 15288 — *Systems and Software Engineering — System Life Cycle Processes*.
- ISO/IEC/IEEE 29148 — *Systems and Software Engineering — Life Cycle Processes — Requirements Engineering*.
- ISO 14971 — *Medical Devices — Application of Risk Management to Medical Devices* (engineering principles relevant to structured risk management).
- Relevant software engineering literature concerning requirements engineering, architectural separation, engineering traceability and runtime verification.

Appendix C – Engineering Design Decisions

C.1 Purpose

Throughout the development of the ASVH Engineering Methodology, a number of engineering decisions were made to ensure that the resulting methodology remained deterministic, implementation independent and suitable for independent Runtime Authority implementations.

This appendix records the principal engineering decisions that influenced the structure of the methodology and explains the rationale supporting each decision. The appendix is intended to provide transparency into the engineering philosophy underpinning the methodology rather than to define additional engineering requirements.

The decisions documented below should therefore be regarded as engineering design rationale rather than normative requirements.

C.2 Methodology Design Principles

The methodology was developed according to five overarching engineering principles:

- preserve the intent of published NHS England guidance;
- maintain complete engineering traceability;
- ensure deterministic runtime evaluation;
- separate engineering methodology from implementation;
- preserve implementation independence.

These principles informed every stage of the engineering lifecycle described throughout this publication.

C.3 Principal Engineering Decisions

Engineering Decision	Engineering Rationale
Separate Operational Requirements from Runtime Requirements	Operational expectations describe organisational behaviour, whereas Runtime Requirements define execution-time conditions required for deterministic evaluation.
Treat Runtime Context as a distinct engineering artefact	Runtime evaluation requires a complete and observable operational state independent of workflow history.
Derive Runtime Rules from Runtime Requirements rather than directly from guidance	Preserves structured engineering transformation and improves traceability.
Evaluate Runtime Rules independently	Simplifies engineering review, testing, maintenance and explainability.
Preserve immutable Runtime Context during evaluation	Ensures deterministic Runtime Authority behaviour and reproducible outcomes.
Extend engineering traceability to Authority Receipts	Completes the engineering evidence chain from published guidance to runtime execution.
Identify the Execution Bind Point through engineering analysis	Ensures Runtime Authority evaluates only when complete Runtime Context exists and execution remains preventable.
Separate methodology from architecture	Preserves implementation independence and enables alternative architectural realisations.
Separate architecture from implementation	Prevents engineering methodology becoming coupled to any particular technology platform or software design.

C.4 Engineering Trade-offs

Several alternative approaches were considered during development of the methodology.

The methodology deliberately avoids:

- embedding Runtime Authority within workflow implementations;
- deriving Runtime Rules directly from narrative guidance;
- relying upon implementation-specific workflow state;

- combining multiple engineering responsibilities within a single artefact;
- coupling Runtime Authority behaviour to individual software platforms.

Instead, the methodology adopts progressive engineering refinement in which each engineering activity produces clearly defined outputs that become the documented inputs to the subsequent stage.

This approach increases engineering transparency, simplifies independent review and preserves implementation independence across multiple deployment environments.

C.5 Relationship to Companion Publications

The engineering decisions documented within this appendix are reflected throughout the ASVH publication series.

Publication	Relationship
ASVH-METH-001	Defines the engineering methodology and design decisions.
ASVH-TRACE-001	Preserves the engineering relationships resulting from these decisions.
ASVH-SPEC-001	Realises the Runtime Rules derived through the methodology.
ASVH-ARCH-001	Implements the architectural principles resulting from the methodology.
ASVH-001	Demonstrates the practical application of the engineering decisions within an executable Runtime Authority.

Together, these publications demonstrate how the engineering decisions documented within this appendix are progressively realised throughout the ASVH engineering framework.

C.6 Closing Observation

The engineering decisions documented within this appendix were made to support the principal objectives of the ASVH Engineering Methodology: deterministic behaviour,

implementation independence, engineering traceability and evidential accountability. Collectively, they explain the rationale for the methodology's structure and demonstrate that its organisation reflects deliberate systems engineering choices rather than implementation convenience.

By recording these decisions explicitly, the methodology provides additional transparency for reviewers, implementers and future contributors while preserving a clear distinction between engineering rationale and normative engineering requirements.

Appendix D – Relationship to the ASVH Engineering Framework

D.1 Purpose

The ASVH Engineering Methodology defines the engineering process through which published NHS England guidance is transformed into deterministic Runtime Authority behaviour. While this publication describes the derivation of the engineering artefacts, the complete ASVH engineering framework is realised through a coordinated series of companion publications, each addressing a distinct engineering responsibility.

This appendix explains the relationship between those publications and illustrates how they collectively form a complete implementation-independent engineering framework for deterministic runtime verification.

D.2 The ASVH Engineering Framework

The ASVH publication series has been deliberately structured so that each publication addresses a single engineering concern while remaining explicitly connected through documented engineering artefacts and traceability relationships.

Rather than duplicating information across multiple documents, each publication builds upon the outputs of the preceding engineering activity.

This separation preserves clarity, simplifies maintenance and enables independent review of individual engineering disciplines while maintaining complete engineering provenance across the framework.

D.3 Publication Responsibilities

Publication	Primary Responsibility	Principal Engineering Output
ASVH-METH-001	Engineering methodology	Runtime Requirements Engineering Process
ASVH-TRACE-001	Engineering traceability	Guidance-to-Implementation Traceability Matrix
ASVH-SPEC-001	Runtime specification	Runtime Context and Runtime Rules
ASVH-ARCH-001	Reference architecture	Implementation-independent Runtime Architecture
ASVH-001	Reference implementation	Executable Runtime Authority and Authority Receipts

Each publication represents a distinct engineering view while contributing to a single, coherent engineering framework.

D.4 Engineering Information Flow

The publications exchange engineering artefacts rather than implementation-specific information.

The engineering information flow proceeds as follows:

1. **ASVH-METH-001** defines the engineering methodology used to derive Runtime Requirements, Runtime Context and Runtime Rules.
2. **ASVH-TRACE-001** records the engineering relationships linking those artefacts to their originating NHS England guidance.
3. **ASVH-SPEC-001** formalises the Runtime Context definitions and deterministic Runtime Rules produced by the methodology.
4. **ASVH-ARCH-001** identifies the architectural location at which the specified Runtime Rules are evaluated by the Runtime Authority.
5. **ASVH-001** demonstrates the methodology through an executable Runtime Authority implementation that generates Authority Receipts and supporting engineering evidence.

This progression preserves a continuous engineering chain from published guidance through to deterministic runtime verification.

D.5 Separation of Responsibilities

The ASVH publication framework intentionally separates engineering responsibilities.

Engineering Activity	Publication Responsible
Guidance Analysis	ASVH-METH-001
Runtime Requirements Engineering	ASVH-METH-001
Engineering Traceability	ASVH-TRACE-001
Runtime Rule Definition	ASVH-SPEC-001
Runtime Architecture	ASVH-ARCH-001
Runtime Verification	ASVH-001
Authority Receipt Generation	ASVH-001

This separation ensures that modifications within one engineering discipline do not unnecessarily affect the responsibilities of another publication.

D.6 Engineering Framework Characteristics

The coordinated publication framework exhibits several important engineering characteristics:

- **Separation of Concerns** – Each publication addresses a single engineering responsibility.
- **Implementation Independence** – Engineering artefacts remain independent of software technology and deployment architecture.
- **Complete Traceability** – Every engineering artefact remains linked to its originating guidance.
- **Deterministic Engineering** – Runtime behaviour is derived through a structured and repeatable engineering methodology.
- **Evidential Accountability** – Authority Receipts provide objective engineering evidence supporting every Runtime Authority decision.

Collectively, these characteristics allow the ASVH publications to function as an integrated engineering framework while preserving the independence of each individual document.

D.7 Closing Observation

The ASVH Engineering Methodology is intended to operate as the foundation of a broader engineering framework rather than as an isolated publication. The companion documents progressively realise the engineering artefacts produced by the methodology, transforming them into traceability models, runtime specifications, reference architectures and executable implementations.

Viewed together, the ASVH publication series demonstrates a complete engineering pathway from published NHS England guidance through to deterministic Runtime Authority behaviour supported by objective runtime evidence. Each publication contributes a distinct engineering perspective, yet all are connected through the shared principles of implementation independence, deterministic evaluation, complete engineering traceability and evidential accountability.

Appendix E – Engineering Terminology

E.1 Purpose

The ASVH Engineering Methodology introduces several engineering concepts that are related but intentionally distinct. Although these terms are used consistently throughout the publication, understanding the relationships between them is essential to understanding the engineering transformation described by the methodology.

This appendix explains the principal engineering artefacts introduced throughout the publication and describes how they progressively transform published NHS England guidance into deterministic Runtime Authority behaviour.

The appendix is explanatory rather than normative and should be read alongside the glossary.

E.2 Engineering Progression

The methodology transforms engineering artefacts through a sequence of progressively more precise representations.

Engineering Artefact	Purpose
NHS England Guidance	Published operational expectations.
Operational Statement	Engineering interpretation of guidance.
Operational Requirement	Measurable operational behaviour.
Runtime Requirement	Execution-time engineering condition.
Runtime Context	Observable operational information presented immediately before execution.
Runtime Rule	Deterministic engineering control evaluating a single runtime condition.
Runtime Authority Decision	Aggregated deterministic execution outcome.
Authority Receipt	Engineering evidence preserving runtime evaluation and decision provenance.

Each engineering artefact represents a refinement of the preceding artefact while preserving complete engineering traceability.

E.3 Relationship Between Engineering Artefacts

The engineering artefacts introduced by the methodology should not be regarded as independent concepts.

Instead, each artefact contributes a distinct engineering responsibility within the overall transformation process.

For example:

- Operational Requirements describe expected organisational behaviour.
- Runtime Requirements identify execution-relevant operational conditions.
- Runtime Context captures the observable information required to evaluate those conditions.
- Runtime Rules express the deterministic logic applied by the Runtime Authority.
- Authority Receipts preserve objective evidence of runtime evaluation.

Together, these artefacts establish a complete engineering pathway from published operational guidance to deterministic runtime verification.

E.4 Relationship to Companion Publications

Each engineering artefact is progressively realised within the ASVH publication series.

Engineering Artefact	Primary Publication
Engineering Methodology	ASVH-METH-001
Engineering Traceability	ASVH-TRACE-001
Runtime Rules	ASVH-SPEC-001
Runtime Architecture	ASVH-ARCH-001
Executable Runtime Authority	ASVH-001

The publications therefore describe complementary engineering perspectives rather than independent engineering models.

E.5 Engineering Lifecycle Summary

The methodology can be summarised as a progressive engineering transformation.

Published NHS England Guidance



Operational Statements



Operational Requirements



Runtime Requirements



Runtime Context



Runtime Rules



Runtime Authority



Every transition within this sequence is governed by the engineering methodology described throughout this publication and preserves complete engineering provenance.

E.6 Closing Observation

The terminology introduced throughout the ASVH Engineering Methodology reflects a deliberate systems engineering approach in which each engineering artefact has a distinct purpose, clearly defined boundaries and explicit relationships to the artefacts that precede and follow it.

Understanding these relationships enables the methodology to be interpreted consistently while preserving implementation independence, deterministic behaviour, engineering traceability and evidential accountability across the wider ASVH engineering framework.

Bibliography

Purpose

The ASVH Engineering Methodology has been developed by applying established systems engineering, requirements engineering, software architecture and verification principles to the specific challenge of transforming published NHS England operational guidance into deterministic Runtime Authority behaviour.

The publications listed below acknowledge the principal standards, guidance and engineering disciplines that informed the development of the methodology. They provide context for the engineering approach adopted throughout this publication but should not be interpreted as defining the ASVH methodology itself.

The principal contribution of this publication lies in the structured engineering process through which published operational guidance is transformed into implementation-independent Runtime Requirements, Runtime Context, Runtime Rules and deterministic Runtime Authority behaviour immediately before institutional consequence formation.

NHS Guidance

NHS England

Guidance on the Use of AI-enabled Ambient Scribing Products in Health and Care Settings.

Primary operational guidance used throughout the ASVH Engineering Methodology.

Systems Engineering

International Council on Systems Engineering (INCOSE)

INCOSE Systems Engineering Handbook.

Provides the systems engineering principles underpinning lifecycle management, engineering decomposition and structured systems development.

ISO/IEC/IEEE 15288

Systems and Software Engineering — System Life Cycle Processes.

Provides internationally recognised lifecycle processes informing the structured engineering approach adopted by the methodology.

ISO/IEC/IEEE 29148

Systems and Software Engineering — Life Cycle Processes — Requirements Engineering.

Provides established requirements engineering concepts supporting the transformation of operational guidance into measurable engineering requirements.

Software Architecture

ISO/IEC/IEEE 42010

Systems and Software Engineering — Architecture Description.

Provides architectural description principles supporting the separation of engineering methodology from architectural realisation.

ISO/IEC 25010

Systems and Software Quality Models.

Provides software quality concepts relevant to deterministic behaviour, maintainability and engineering quality attributes.

Clinical Safety

DCB0129

Clinical Risk Management: Its Application in the Manufacture of Health IT Systems.

Recognised NHS clinical safety standard relevant to healthcare software engineering.

DCB0160

Clinical Risk Management: Its Application in the Deployment and Use of Health IT Systems.

Recognised NHS clinical safety standard governing deployment and operational use of healthcare information systems.

Requirements, Verification and Traceability

The ASVH Engineering Methodology applies established engineering principles relating to:

- requirements engineering;
- engineering traceability;
- systems verification;
- software verification;
- deterministic engineering; and
- evidential accountability.

These engineering disciplines provide the methodological foundation upon which Runtime Requirement Engineering, Runtime Context Engineering and Runtime Rule Derivation have been developed.

Companion Publications

The engineering artefacts described throughout this publication are progressively realised through the companion ASVH publications.

- **ASVH-METH-001** — *Engineering Methodology*
- **ASVH-TRACE-001** — *Runtime Traceability Matrix*
- **ASVH-SPEC-001** — *Runtime Authority Specification*
- **ASVH-ARCH-001** — *Reference Architecture*
- **ASVH-001** — *Reference Implementation*

Collectively, these publications define a complete engineering framework supporting deterministic Runtime Authority behaviour derived from published NHS England operational guidance.

Closing Observation

The references listed within this bibliography acknowledge the established engineering disciplines that informed the development of the ASVH Engineering Methodology. The methodology does not seek to replace these established practices. Instead, it combines and extends them to address the specific engineering challenge of transforming narrative operational guidance into deterministic, implementation-independent Runtime Authority behaviour supported by complete engineering traceability and objective runtime evidence.

Document Identifier

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Status

Public Release

Related Publications

- ASVH-AUTH-001
- ASVH-SPEC-001
- ASVH-TRACE-001
- ASVH-001 (Reference Implementation Report)